

Error Codes Directory

All Robots



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Contents

1. Error Codes CB3 and Earlier Models	11
1.1. C0 No error	11
1.2. C1 Outbuffer overflow	11
1.3. C2 Inbuffer overflow	11
1.4. C3 Processor overloaded	11
1.5. C4 Communication issue	11
1.6. C5 Heavy processor load warning	17
1.7. C10 Controller communication issue	17
1.8. C11 Bad CRC	17
1.9. C12 Unknown message error	17
1.10. C14 Debug message	17
1.11. C17 Communication error between Safety Control Board and Motherboard	18
1.12. C25 Motor Encoder index missing	18
1.13. C26 Motor Encoder index drift detected	18
1.14. C27 Calibration data is invalid or does not exist, selftest is needed!	18
1.15. C29 Online Calibration data checksum failed	18
1.16. C30 Master received data from too many joints	18
1.17. C31 Caught wrong message (not from master)	18
1.18. C32 Flash write verify failed	18
1.19. C33 Calibration flash checksum failed	18
1.20. C34 Program flash checksum failed	18
1.21. C35 Joint ID is undefined	19
1.22. C36 Illegal bootloader command	19
1.23. C37 Inbuffer parse error	19
1.24. C38 Online RAM test failed	19
1.25. C39 Logic and Temporal Monitoring Fault	19
1.26. C40 AD-Converter hit high limit joint	20
1.27. C41 RC Oscillator Trim register hit high limit	20
1.28. C42 RC Oscillator Trim register hit low limit	20
1.29. C43 Change in invariant memory detected	20
1.30. C44 CRC check failure on primary bus	20
1.31. C45 AD-Converter error	21
1.32. C46 Loose gearbox or bad encoder mounting	21
1.33. C47 AD-Converter hit low limit	21
1.34. C49 RS485 receive warning	21
1.35. C50 Robot powerup issue	21
1.36. C51 CRC check failure on secondary bus	23
1.37. C53 IO overcurrent detected	23
1.38. C55 Safety system error	24



1.39. C56 Overvoltage shutdown	25
1.40. C57 Brake release failure	25
1.41. C58 Motor encoder not calibrated	25
1.42. C59 Overcurrent shutdown	25
1.43. C60 Energy surplus shutdown	26
1.44. C61 Idle power consumption to high	26
1.45. C62 Thermal issue	26
1.46. C63 Motor test failed in step {unsigned}.	26
1.47. C64	26
1.48. C65 PSU voltage to high	26
1.49. C68 SPI error	26
1.50. C70 Close to gearbox shear limit	26
1.51. C71 Startup check error	26
1.52. C72 Power Supply Unit failure	28
1.53. C73 Brake test failed during selftest, check brakepin	29
1.54. C74 Joint encoder warning	29
1.55. C75 Joint encoder error	29
1.56. C76 Joint encoder communication CRC issue	30
1.57. C77 Sudden position change detected on the joint-encoder	30
1.58. C78 Large sudden position change detected on the joint-encoder	30
1.59. C85 Motor encoder error	30
1.60. C100 Robot changed mode	30
1.61. C101 Real Robot Connected	30
1.62. C102 Real Robot not connected - Simulating Robot	30
1.63. C103 Communication issue	31
1.64. C104 Error=Empty command sent to robot	31
1.65. C111 Something is pulling the robot	31
1.66. C115 Unknown robot type	31
1.67. C116 Realtime part warning	31
1.68. C117 Restart SCB failed	31
1.69. C150 Position close to joint limits	31
1.70. C151 Tool orientation close to limits	31
1.71. C152 Position close to safety plane limits	32
1.72. C153 Position deviates from path	32
1.73. C154 Position in singularity	33
1.74. C155 Robot cannot maintain its position, check if payload is correct	33
1.75. C156 Wrong payload or mounting detected, or something is pushing the robot when entering Freedrive mode	33
1.76. C157 Collision detected by joint	33
1.77. C158 Collision detected by joint	34
1.78. C159 Position deviates from path	35
1.79. C160 The robot was powered off last time due to a joint position disagreement	35

1.80. C161 Large movement of the robot detected while it was powered off. The joints were moved while it was powered off, or the encoders do not function	36
1.81. C162 The protective stop was likely caused by incorrectly specified payload mass and/or center of gravity.	36
1.82. C163 More than 50 Protective Stops are detected on the same joint within 8 hours of operation.	36
1.83. C164 Wrist position close to safety plane limits	37
1.84. C171 Issue with blends	37
1.85. C172 Illegal control mode	38
1.86. C173 Robot motion causes too high joint torques	38
1.87. C174 Robot motion causes too high jump in joint torques	39
1.88. C184 Joint self test not received by controller	39
1.89. C185	39
1.90. C186	39
1.91. C187 Temperature sensor test failed	39
1.92. C190 Joint failed during selftest	40
1.93. C191 Safety system violation	40
1.94. C192 Safety system fault	42
1.95. C193 One of the nodes is in fault mode	45
1.96. C194 One of the nodes is not booted or not present	46
1.97. C195 Conveyor speed too high	47
1.98. C196 MoveP speed too high	47
1.99. C197 Blend overlap warning	48
1.100. C200 Safety Control Board hardware error	48
1.101. C201 Setup of Safety Control Board failed	51
1.102. C202 SCE configuration was illegal, after applying tolerances	51
1.103. C203 PolyScope detected a mismatch between the shown and (to be) applied safety parameters	52
1.104. C204 Path sanity check failed	52
1.105. C205 Target speed does not match target position	52
1.106. C206 Sanity check failed	52
1.107. C207 Fieldbus input disconnected	53
1.108. C208 Debug Assertion failed	53
1.109. C209 A protective stop was triggered (for test purposes only)	53
1.110. C210 Socket is read-only when the robot is in local (Teach pendant) control	53
1.111. C211 Operational mode changed	53
1.112. C212 Name conflict in loaded program	53
1.113. C213 No Kinematic Calibration found (calibration.conf file is either corrupt or missing)	53
1.114. C214 Kinematic Calibration for the robot does not match the joint(s)	54
1.115. C215 Kinematic Calibration does not match the robot	54
1.116. C216 The offset of the joint has changed	54
1.117. C217 White space detected at the beginning of a string token at line {unsigned}	54
1.118. C218 A thread used a lot of time	54
1.119. C219 Path Offset	54
1.120. C220 Kinematic Calibration	55



1.121. C221 GUI Communication	55
1.122. C222 Frame Tracking	56
1.123. C223 Flexible EtherNet/IP Fieldbus input (custom instance) disconnected	57
1.124. C224 {string}	57
1.125. C225 Logic Program '{string}' is not in running state.	57
1.126. C226 A fault was triggered (for test purposes only)	57
2. Error Codes e-Series and Later Models	58
2.1. C259 Filesystem-related issue	58
2.2. C260 Brake Release - old procedure	58
2.3. C261 Temperature Sensor	59
2.4. C262 Communication	62
2.5. C263 Motor Encoder	63
2.6. C264 Task Manager	64
2.7. C265 Joint Encoder	66
2.8. C266 Self-test	68
2.9. C267 Bootloader error	68
2.10. C268 Special Command	69
2.11. C269 Transceiver - deprecated	69
2.12. C271 Low level real-time thread	70
2.13. C272 Missing calibration	70
2.14. C273 Cross monitoring	70
2.15. C274 Control box fan error	78
2.16. C275 Brake Pin	79
2.17. C276 Uart	79
2.18. C277 Memory	79
2.19. C278 Servo	80
2.20. C279 Flash	80
2.21. C280 Real-time error	81
2.22. C281 Robot State Machine	81
2.23. C283 Safety system	83
2.24. C284 Brake Release	102
2.25. C285 Joint Keep-Alive System	103
2.26. C286 Motor Controller	103
2.27. C287 Saved files	104
2.28. C289 Tool Connector	104
2.29. C292 Online RAM test	106
2.30. C294 ADC	106
2.31. C295 PCB	107
2.32. C296 Start up check	107
2.33. C297 Joint validation	110
2.34. C298 Hand protection	111
2.35. C299 Tool communication	111

2.36. C301 Safety message monitor	112
2.37. C302 Tool Configuration	112
2.38. C303 System status	112
2.39. C304 Self monitoring	113
2.40. C305 Robot Power Control	114
2.41. C306 Joint	115
2.42. C309 Keep-Alive	117
2.43. C310 Joint Temperature Manager	118
2.44. C312 Data validation	118
2.45. C314 SPI IO	119
2.46. C315 Watchdog	120
2.47. C316 MCU	121
2.48. C328 Transceiver miscellaneous	122
2.49. C329 AXI STREAM	122
2.50. C330 IMMI IO	122
2.51. C331 Friction model	123
2.52. C332 Servo configuration	124
2.53. C333 File message	124
2.54. C334 Robot deviated from constrained axes while in Constrained FreeDrive.	125
2.55. C337 Control parameters	125
2.56. C338 PROFIsafe	125
2.57. C339 Cross communication	127
2.58. C340 Energy Monitoring	127
2.59. C341 Motor encoder	128
2.60. C342 Motor Parameters	132
2.61. C343 Joint Configuration	132
2.62. C347 Network Map	135
2.63. C348 External Axes Bus	136
2.64. C349 Base light	136
2.65. C350 Failed brake system	137
2.66. C351 Robot configuration	138
2.67. C352 Backdrive	139
2.68. C353 IMU	140
2.69. C354 Initialization Error	142
2.70. C355 Safety API	144
2.71. C357 Too high static load on {deviceName}	146
2.72. C359 Report system	147
2.73. C361 Gear torsion	147
2.74. C362 State Machine	147
2.75. C363 Tool Accelerometer	148
2.76. C364 Robot validation	148
2.77. C365 Teach Pendant RS485 communication	149



2.78. C366 Tool panel interface	149
2.79. C367 Power Control Board	150
2.80. C368 Power Supply Unit	151
2.81. C370 Ethernet MDIO	151
2.82. C371 End of Arm Ethernet	151
2.83. C373 Power Interface Board	151
2.84. C375 Motor encoder	152
2.85. C377 Charging system	152
2.86. C400 Elbow position close to safety plane limits	152
2.87. C401 Exceeding user safety settings for stopping time	152
2.88. C402 Exceeding user safety settings for stopping distance	152
2.89. C403 Danger of clamping between the lower arm and tool flange of the robot	153
2.90. C404 Unexpected behavior	153
2.91. C450 Force-Torque sensor	153
2.92. C451 Tool Power Sensor	154
2.93. C453 SCB IMU	154
2.94. C500 Self-test step	155
2.95. C526 Node ID	155
2.96. C710 ROM Test	156
2.97. C720 LVD (low voltage detection)	156
2.98. C740 Hardware monitoring	157
2.99. C741 Hardware general information	163
2.100. C742 Control Box temperature	163
2.101. C743 External Axes	163
2.102. C746 Cable Extender	165
2.103. C748 Security	166
2.104. C749 Digital output	166

1. Error Codes CB3 and Earlier Models

Error codes with numbers 1 to 255 apply only to CB3 and earlier robot models.

1.1. C0 No error

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.2. C1 Outbuffer overflow

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C1A1 Buffer of stored warnings overflowed

C1A2 Outbuffer to RS485 overflowed (problem with Controller message)

1.3. C2 Inbuffer overflow

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.4. C3 Processor overloaded

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.5. C4 Communication issue

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C4A1 Lost communication with Controller

EXPLANATION

Communication was lost between the Safety Control Board and the Motherboard

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check Ethernet cable between Safety Control Board and Motherboard, check that a script or UR+ software is not overloading the communication between the Safety Control Board and Motherboard, (B) Conduct a complete rebooting sequence, (C) Update the software

C4A2 Lost communication with Safety Control Board A uP

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check TCP/IP connection between Motherboard and Safety Control Board, (B) Conduct a complete rebooting sequence, (C) Exchange Safety Control Board

C4A3 Communication with Safety Control Board B uP lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check TCP/IP connection between Motherboard and Safety Control Board, (B) Conduct a complete rebooting sequence, (C) Exchange Safety Control Board



C4A4 Communication with primary Teach Pendant uP lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check RS485-12V connection between Motherboard and Teach Pendant, (B) Conduct a complete rebooting sequence, (C) Exchange Teach Pendant

C4A5 Communication with secondary Teach Pendant uP lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check RS485-12V connection between Motherboard and Teach Pendant, (B) Conduct a complete rebooting sequence, (C) Exchange Teach Pendant

C4A6 Communication with primary EUROMAP67 uP lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check Euromap67 connection between Motherboard and Teach Pendant, (B) Conduct a complete rebooting sequence, (C) Exchange Teach Pendant

C4A7 Communication with secondary EUROMAP67 uP lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check Euromap67 connection between Motherboard and Teach Pendant, (B) Conduct a complete rebooting sequence, (C) Exchange Teach Pendant

C4A8 Primary EUROMAP67 uP present, but euromap67 is disabled

EXPLANATION

Incorrect safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update the miscellaneous settings in the Safety Configuration, (B) Conduct a complete rebooting sequence

C4A9 Secondary EUROMAP67 uP present, but euromap67 is disabled

EXPLANATION

Incorrect safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update the miscellaneous settings in the Safety Configuration, (B) Conduct a complete rebooting sequence

C4A10 Primary Teach Pendant present, but Teach Pendant safety is disabled

EXPLANATION

Incorrect safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update the miscellaneous settings in the Safety Configuration, (B) Conduct a complete rebooting sequence

C4A11 Secondary Teach Pendant uP present, Teach Pendant safety is disabled

EXPLANATION

Incorrect safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update the miscellaneous settings in the Safety Configuration, (B) Conduct a complete rebooting sequence

C4A12 Communication with joint 0 lost

EXPLANATION

More than 1 package lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A13 Communication with joint 1 lost

EXPLANATION

More than 1 package lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A14 Communication with joint 2 lost

EXPLANATION

More than 1 package lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A15 Communication with joint 3 lost

EXPLANATION

More than 1 package lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A16 Communication with joint 4 lost

EXPLANATION

More than 1 package lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly (B) Conduct a complete rebooting sequence

C4A17 Communication with joint 5 lost

EXPLANATION

More than 1 package lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A18 Communication with tool lost

EXPLANATION

More than 1 package lost

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly (B) Conduct a complete rebooting sequence

C4A65 Lost package from Primary Teach Pendant

EXPLANATION

1 package lost

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A66 Lost package from Secondary Teach Pendant

EXPLANATION

1 package lost

**SUGGESTION**

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A67 Lost package from Primary Euromap67**EXPLANATION**

1 package lost

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A68 Lost package from Secondary Euromap67**EXPLANATION**

1 package lost

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A69 Lost package from Secondary Masterboard**EXPLANATION**

1 package lost

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A70 Lost package from joint 0**EXPLANATION**

Serial communication problem with one or more joints

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A71 Lost package from joint 1**EXPLANATION**

Serial communication problem with one or more joints

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A72 Lost package from joint 2**EXPLANATION**

Serial communication problem with one or more joints

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A73 Lost package from joint 3**EXPLANATION**

Serial communication problem with one or more joints

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A74 Lost package from joint 4

EXPLANATION

Serial communication problem with one or more joints

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A75 Lost package from joint 5**EXPLANATION**

Serial communication problem with one or more joints

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A76 Lost package from tool**EXPLANATION**

Serial communication problem with one or more joints

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A77 Lost package from uPA to joints**EXPLANATION**

1 package lost

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A78 Lost package from uPA to teach pendant**EXPLANATION**

1 package lost

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A79 Lost package from uPA to uPB**EXPLANATION**

1 package lost

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A80 Lost package from uPB**EXPLANATION**

1 package lost

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A81 Packet counter disagreement in packet from Primary Screen**EXPLANATION**

Safety processor 1 in Teach Pendant has a packet disagreement

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence



C4A82 Packet counter disagreement in packet from Secondary Screen

EXPLANATION

Safety processor 2 in Teach Pendant has a packet disagreement

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A83 Packet counter disagreement in packet from Primary Euromap67

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A84 Packet counter disagreement in packet from Secondary Euromap67

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A85 Packet counter disagreement in packet from Safety Control Board B

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A86 Packet counter disagreement in packet from joint 0

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A87 Packet counter disagreement in packet from joint 1

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A88 Packet counter disagreement in packet from joint 2

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A89 Packet counter disagreement in packet from joint 3

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A90 Packet counter disagreement in packet from joint 4

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A91 Packet counter disagreement in packet from joint 5

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A92 Packet counter disagreement in packet from tool

SUGGESTION

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A93 Packet counter disagreement in packet from processor A to joints**SUGGESTION**

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A94 Packet counter disagreement in packet from processor A to B**SUGGESTION**

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A95 Packet counter disagreement in packet from processor A to Teach Pendant and EUROMAP**SUGGESTION**

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

C4A100 Communication lost due to Packet counter disagreements**SUGGESTION**

If this happens often, try the following actions to see which resolves the issue: (A) Verify the communication cables are connected properly, (B) Conduct a complete rebooting sequence

1.6. C5 Heavy processor load warning

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.7. C10 Controller communication issue

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C10A1 Lost packet from Controller**C10A101 Controller packet received too early****C10A102 Packet counter does not match****C10A103 Controller is sending packets too often**

1.8. C11 Bad CRC

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.9. C12 Unknown message error

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.10. C14 Debug message

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C14A1 {float}



C14A2 {signed}

C14A3 {unsigned}

1.11. C17 Communication error between Safety Control Board and Motherboard

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.12. C25 Motor Encoder index missing

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.13. C26 Motor Encoder index drift detected

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.14. C27 Calibration data is invalid or does not exist, selftest is needed!

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.15. C29 Online Calibration data checksum failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.16. C30 Master received data from too many joints

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.17. C31 Caught wrong message (not from master)

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.18. C32 Flash write verify failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.19. C33 Calibration flash checksum failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.20. C34 Program flash checksum failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C34A0 Program flash checksum failed during bootloading

C34A1 Program flash checksum failed at runtime

1.21. C35 Joint ID is undefined

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.22. C36 Illegal bootloader command

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.23. C37 Inbuffer parse error

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.24. C38 Online RAM test failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C38A1 Data-bus test failed

C38A2 Address-bus stuck-high test failed

C38A3 Address-bus stuck-low test failed

C38A4 Address-bus shorted test failed

C38A5 Memory-cell test failed

1.25. C39 Logic and Temporal Monitoring Fault

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C39A1 Max current deviation failure

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice, replace joint

C39A2 Max joint-encoder speed exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice, replace joint

C39A3 Max motor-encoder speed exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice, replace joint

C39A4 Illegal state change in joint detected



C39A5 A timing issue occurred during startup.

EXPLANATION

Too fast state change in joint detected

SUGGESTION

Conduct a complete rebooting sequence

C39A6 5V regulator voltage too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice, replace joint

C39A7 5V regulator voltage too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice, replace joint

C39A100 Watchpoint fault: ADC task timeout

C39A101 Watchpoint fault: Motor-Control task timeout

C39A102 Watchpoint fault: Motor-encoder task timeout

C39A103 Watchpoint fault: Joint-encoder task timeout

C39A104 Watchpoint fault: Communication task timeout

C39A105 Watchpoint fault: RAM-test task timeout

C39A106 Watchpoint fault: CalVal-test task timeout

C39A107 Watchpoint fault: ROM-test task timeout

1.26. C40 AD-Converter hit high limit joint

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.27. C41 RC Oscillator Trim register hit high limit

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.28. C42 RC Oscillator Trim register hit low limit

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.29. C43 Change in invariant memory detected

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C43A1 Current sensor gain

1.30. C44 CRC check failure on primary bus

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C44A0 Base

C44A1 Shoulder

C44A2 Elbow

C44A3 Wrist 1

C44A4 Wrist 2

C44A5 Wrist 3

C44A6 Tool

C44A80 CRC Check failure on primary bus.

EXPLANATION

Most likely an interference on the communication bus.

1.31. C45 AD-Converter error

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.32. C46 Loose gearbox or bad encoder mounting

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.33. C47 AD-Converter hit low limit

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.34. C49 RS485 receive warning

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C49A200 Secondary RS485 bus is down

EXPLANATION

Bus for: Teach Pendant, Processor A and Processor B on SCB

SUGGESTION

Check TCP/IP-12V cable to Teach Pendant

1.35. C50 Robot powerup issue

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C50A1 Voltage detected at 24V rail before startup

C50A2 Voltage present at unpowered robot

C50A5 Powersupply voltage too low

SUGGESTION

Check 48 V cable between power supply and SCB

C50A6 Powersupply voltage too high

C50A11 Voltage not detected at 24V rail after startup

C50A15 Warning, waiting for SafetySYS2



C50A16 The Teach Pendant does not respond

EXPLANATION

Loose wire or incorrect safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the Teach Pendant cable and connections, (B) Check the settings in the miscellaneous tab in the Safety menu

C50A17 The Euromap67 interface does not respond

EXPLANATION

Loose wire or incorrect safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the Euromap67 cable and connections, (B) Check the settings in the miscellaneous tab in the Safety menu

C50A18 Warning, waiting for SafetySYS1

EXPLANATION

SafetySYS1 = Processor A on Safety Control Board

C50A19 Warning, Waiting for a valid "euromap67 activated" status bit from secondary Safety Control Board

C50A20 5V, 3V3 or ADC error (5V too high)

C50A21 5V, 3V3 or ADC error (5V too low)

C50A22 Robot current sensor reading too high

C50A23 Robot current sensor reading too low

C50A24 48V not present (Check internal connection)

C50A25 Robot voltage present at 48V PSU powereup

C50A26 Voltage present on unpowered 48V power supply

C50A27 12V, 3V3 or ADC error (12V too high)

C50A28 12V, 3V3 or ADC error (12V too low)

C50A29 Analog I/O error (-12V too high)

C50A30 Analog I/O error (-12V too low)

C50A31 The other safetySYS do not initialize

C50A40 Wrong voltage from PSU1

C50A41 Wrong voltage from PSU2

C50A42 Voltage will not disappear from PSU

C50A43 Warning, waiting for CB2 type answer from primary processor

C50A50 Processor A 3.3V supply voltage out of bounds

C50A51 Robot voltage below threshold

C50A52 Robot voltage above threshold

C50A53 58V generator deviation error

C50A54 5V regulator too low

C50A55 5V regulator too high

C50A56 -4V generator too low

C50A57 -4V generator too high

C50A80 Last CPU reset caused by Low-Power-Reset**C50A81 Last CPU reset caused by Window-Watchdog-Reset****C50A82 Last CPU reset caused by Independent-Watchdog-Reset****C50A83 Last CPU reset caused by Software-Reset****EXPLANATION**

The safety control board was reset on explicit request.

SUGGESTION**C50A84 Last CPU reset caused by External-Pin-Reset****C50A85 Last CPU reset caused by Brown-Out-Reset****C50A99 Wrong software on PCB****C50A100 Cable not connected****SUGGESTION**

Check cable and connections between robot and Control Box

C50A101 Short circuit in robot detected or wrong robot connected to Control Box**SUGGESTION**

Check robot type. Look for short circuit in cable and in Robot Arm

C50A102 Voltage rising too slowly**C50A103 Voltage failed to reach acceptable level****C50A104 The IMMI module does not respond****EXPLANATION**

Missing IMMI module, hardware failure or incorrect safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the IMMI module and connections, (B) Check the settings in the Hardware tab in the Safety section of the installation

1.36. C51 CRC check failure on secondary bus

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C51A0 Processor B**C51A1 Primary screen processor****EXPLANATION**

CRC check failure on safety processor 1 in Teach Pendant

C51A2 Secondary screen processor**EXPLANATION**

CRC check failure on safety processor 2 in Teach Pendant

C51A3 Primary E67**C51A4 Secondary E67**

1.37. C53 IO overcurrent detected

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C53A1 , max is 800mA



C53A2 , max is 600mA

1.38. C55 Safety system error

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C55A23 Safety relay error (minus connection)

EXPLANATION

Current distributor error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check cable from Safety Control Board to Current distributor or 48V Power supply and Current distributor for issues, (B) Conduct a complete rebooting sequence

C55A24 Safety relay error (plus connection)

EXPLANATION

Current distributor error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check cable from Safety Control Board to Current distributor or 48V Power supply and Current distributor for issues, (B) Conduct a complete rebooting sequence

C55A33 Safety relay error (a relay is stuck)

EXPLANATION

Current distributor error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check cable from Safety Control Board to Current distributor or 48V Power supply and Current distributor for issues (B) Conduct a complete rebooting sequence

C55A34 Safety relay error (relays are not on)

EXPLANATION

Current distributor error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check cable from Safety Control Board to Current distributor or 48V Power supply and Current distributor for issues (B) Conduct a complete rebooting sequence

C55A50 Voltage present at unpowered robot

EXPLANATION

Safety Control Board hardware fault

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C55A51 Voltage will not disappear from robot

EXPLANATION

Safety Control Board hardware fault

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C55A52 5V, 3V3 or ADC error (5V too low)

EXPLANATION

Safety Control Board hardware fault

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C55A53 5V, 3V3 or ADC error (5V too high)**EXPLANATION**

Safety Control Board hardware fault

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C55A90 Bootloader error, robot voltage too low or current too high**C55A91 Bootloader error, robot voltage too high****C55A100 Safety violation****C55A101 Safety Channel Error In Safety Control Board****C55A102 Safety Channel Error In Screen****C55A103 Safety Channel Error In Euromap67 Interface****C55A109 Received fault message from Controller****C55A110 Safety State is changing too often****C55A111 On/Off State is changing too often****C55A112 Robot current sensors readings differ****C55A120 Robot current is too high while emergency stopped****C55A121 Robot current is too high while safeguard stopped**

1.39. C56 Overvoltage shutdown

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.40. C57 Brake release failure

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C57A1 Joint did not move or motor encoder is not functioning**C57A2 Large movement detected during brake release****C57A3 Robot was not able to brake release, see log for details**

1.41. C58 Motor encoder not calibrated

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.42. C59 Overcurrent shutdown

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.



1.43. C60 Energy surplus shutdown

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.44. C61 Idle power consumption too high

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.45. C62 Thermal issue

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C62A1 Joint temperature: High (80(C)

C62A3 Warning: Static load too high

C62A11 Joint temperature: Shut down (85(C)

C62A13 Shutdown: Static load too high

1.46. C63 Motor test failed in step {unsigned}.

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.47. C64

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.48. C65 PSU voltage too high

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.49. C68 SPI error

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.50. C70 Close to gearbox shear limit

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.51. C71 Startup check error

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C71A0 Hardware is size0, wrong firmware at the joint

SUGGESTION

Update firmware

C71A1 Hardware is size1, wrong firmware at the joint

SUGGESTION
Update firmware

C71A2 Hardware is size2, wrong firmware at the joint

SUGGESTION
Update firmware

C71A3 Hardware is size3, wrong firmware at the joint

SUGGESTION
Update firmware

C71A4 Hardware is size4, wrong firmware at the joint

SUGGESTION
Update firmware

C71A5 Invalid hardware revision

C71A6 ADC calibration failed

C71A7 Unknown error result

EXPLANATION
The motor wires are damaged, bad connection in screw terminals or defect PCB

SUGGESTION
Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A8 Motor short circuit to ground or H-bridge problems

EXPLANATION
The motor wires are damaged, bad connection in screw terminals or defect PCB

SUGGESTION
Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A9 Motor indication signal does not work

EXPLANATION
The motor wires are damaged, bad connection in screw terminals or defect PCB

SUGGESTION
Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A10 Phase 1 is unconnected or not working

EXPLANATION
The motor wires are damaged, bad connection in screw terminals or defect PCB

SUGGESTION
Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A11 Phase 2 is unconnected or not working

EXPLANATION
The motor wires are damaged, bad connection in screw terminals or defect PCB

SUGGESTION
Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A12 Phase 3 or multiple phases is unconnected or not working

**EXPLANATION**

The wire is (1) damaged or (2) has been disconnected from the PCB (not likely) or (3) defect PCB

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A50 Current sensor test failed**EXPLANATION**

Sensor reported wrong current when probed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A51 Current sensor test failed**EXPLANATION**

Sensor reported wrong current when probed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A52 Current sensor test failed**EXPLANATION**

Sensors reported different currents when probed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check joint for damaged or loose connections, (B) Replace the joint

C71A101 Wrong firmware on motor encoder

1.52. C72 Power Supply Unit failure

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C72A1 0 PSUs are active**EXPLANATION**

PSU was not able to deliver 48V

SUGGESTION

Check power connection between power supply and Safety Control Board

C72A2 1 PSU active, but we expect 2 (UR10)**EXPLANATION**

PSU was not able to deliver 48V or UR10 flash card in UR5 robot

SUGGESTION

Check power connection between power supply and Safety Control Board and check that the flash card and robot match

C72A3 2 PSUs active, but we expect 1 (UR5)**EXPLANATION**

UR5 flash card in UR10 robot

SUGGESTION

Check that the flash card and robot match

1.53. C73 Brake test failed during selftest, check brakepin

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.54. C74 Joint encoder warning

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C74A1 Invalid decode: Readhead misalignment, ring damaged or external magnetic field present.

SUGGESTION

Check grounding and shielding for EMC problems

C74A2 Speed reading is not valid

C74A4 System error=malfunction or inconsistent calibration detected

C74A8 Supply voltage is out of range

C74A16 Temperature is out of range

C74A32 Signal lost =Misaligned readhead or damaged ring

C74A64 Signal low =Too far from magnetic ring

C74A128 Signal saturation =Too close to magnetic ring

1.55. C75 Joint encoder error

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C75A1 Invalid decode: Readhead misalignment, ring damaged or external magnetic field present.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check grounding and shielding for EMC problems, (C) If this happens more than twice, replace joint

C75A2 Speed reading is not valid

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice, replace joint

C75A4 System error=malfunction or inconsistent calibration detected

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice, replace joint

C75A8 Supply voltage is out of range

C75A16 Temperature is out of range

C75A32 Signal lost =Misaligned readhead or damaged ring

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check grounding and shielding for EMC problems, (C) If this happens more than twice, replace joint

C75A64 Signal low =Too far from magnetic ring**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check grounding and shielding for EMC problems, (C) If this happens more than twice, replace joint

C75A128 Signal saturation =Too close to magnetic ring**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check grounding and shielding for EMC problems, (C) If this happens more than twice, replace joint

C75A200 Position from joint encoder does not change while motor is running

1.56. C76 Joint encoder communication CRC issue

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.57. C77 Sudden position change detected on the joint-encoder

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.58. C78 Large sudden position change detected on the joint-encoder

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.59. C85 Motor encoder error

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C85A200 Position from motor encoder does not change while motor is running

1.60. C100 Robot changed mode

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.61. C101 Real Robot Connected

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.62. C102 Real Robot not connected – Simulating Robot

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.63. C103 Communication issue

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C103A1 Connection to Safety Control Board lost

EXPLANATION

PC did not receive 3 packets in a row

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check that the Ethernet cable between Motherboard and Safety Control Board is connected, (B) Conduct a complete rebooting sequence

C103A2 Package lost from Safety Control Board

C103A3 Ethernet connection initialization with Safety Control Board failed

1.64. C104 Error=Empty command sent to robot

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.65. C111 Something is pulling the robot

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.66. C115 Unknown robot type

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.67. C116 Realtime part warning

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.68. C117 Restart SCB failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.69. C150 Position close to joint limits

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.70. C151 Tool orientation close to limits

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.



1.71. C152 Position close to safety plane limits

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.72. C153 Position deviates from path

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C153A0 Detected by the Base joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C153A1 Detected by the Shoulder joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C153A2 Detected by the Elbow joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C153A3 Detected by the Wrist 1 joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C153A4 Detected by the Wrist 2 joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C153A5 Detected by the Wrist 3 joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

1.73. C154 Position in singularity

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.74. C155 Robot cannot maintain its position, check if payload is correct

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.75. C156 Wrong payload or mounting detected, or something is pushing the robot when entering Freedrive mode

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.76. C157 Collision detected by joint

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C157A0 Detected by the Base joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C157A1 Detected by the Shoulder joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C157A2 Detected by the Elbow joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C157A3 Detected by the Wrist 1 joint.

EXPLANATION

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C157A4 Detected by the Wrist 2 joint.

**EXPLANATION**

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

C157A5 Detected by the Wrist 3 joint.**EXPLANATION**

Robot could not follow the path, either there was a collision, or a setting was incorrect. If protective stops occur frequently, resolve the cause.

SUGGESTION

Make sure no objects are in the path, check payload, center of gravity and acceleration settings.

1.77. C158 Collision detected by joint

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C158A0 Base. The user specified payload is 0kg, please make sure this is correct.**EXPLANATION**

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C158A1 Shoulder. The user specified payload is 0kg, please make sure this is correct.**EXPLANATION**

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C158A2 Elbow. The user specified payload is 0kg, please make sure this is correct.**EXPLANATION**

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C158A3 Wrist 1. The user specified payload is 0kg, please make sure this is correct.**EXPLANATION**

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C158A4 Wrist 2. The user specified payload is 0kg, please make sure this is correct.**EXPLANATION**

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C158A5 Wrist 3. The user specified payload is 0kg, please make sure this is correct.**EXPLANATION**

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

1.78. C159 Position deviates from path

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C159A0 Base. The user specified payload is 0kg, please make sure this is correct.

EXPLANATION

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C159A1 Shoulder. The user specified payload is 0kg, please make sure this is correct.

EXPLANATION

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C159A2 Elbow. The user specified payload is 0kg, please make sure this is correct.

EXPLANATION

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C159A3 Wrist 1. The user specified payload is 0kg, please make sure this is correct.

EXPLANATION

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C159A4 Wrist 2. The user specified payload is 0kg, please make sure this is correct.

EXPLANATION

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

C159A5 Wrist 3. The user specified payload is 0kg, please make sure this is correct.

EXPLANATION

Specifying an incorrect payload mass and/or center of gravity may cause poor robot performance and/or protective stops.

SUGGESTION

Make sure the payload mass and center of gravity are correctly specified.

1.79. C160 The robot was powered off last time due to a joint position disagreement

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.



1.80. C161 Large movement of the robot detected while it was powered off. The joints were moved while it was powered off, or the encoders do not function

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.81. C162 The protective stop was likely caused by incorrectly specified payload mass and/or center of gravity.

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.82. C163 More than 50 Protective Stops are detected on the same joint within 8 hours of operation.

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C163A0 Base joint. Something is wrong in the application. Recurring protective stops should be resolved, ignoring it can void warranty.

EXPLANATION

Frequent protective stops is an indication of either an error in the application, or a hardware failure in a joint.

SUGGESTION

Check that the motions of the robot do not cause collision, check payload, center of gravity and acceleration settings, if these settings are correct, the joint needs repair.

C163A1 Shoulder joint. Something is wrong in the application. Recurring protective stops should be resolved, ignoring it can void warranty.

EXPLANATION

Frequent protective stops is an indication of either an error in the application, or a hardware failure in a joint.

SUGGESTION

Check that the motions of the robot do not cause collision, check payload, center of gravity and acceleration settings, if these settings are correct, the joint needs repair.

C163A2 Elbow joint. Something is wrong in the application. Recurring protective stops should be resolved, ignoring it can void warranty.

EXPLANATION

Frequent protective stops is an indication of either an error in the application, or a hardware failure in a joint.

SUGGESTION

Check that the motions of the robot do not cause collision, check payload, center of gravity and acceleration settings, if these settings are correct, the joint needs repair.

C163A3 Wrist 1 joint. Something is wrong in the application. Recurring protective stops should be resolved, ignoring it can void warranty.

EXPLANATION

Frequent protective stops is an indication of either an error in the application, or a hardware failure in a joint.

SUGGESTION

Check that the motions of the robot do not cause collision, check payload, center of gravity and acceleration settings, if these settings are correct, the joint needs repair.

C163A4 Wrist 2 joint. Something is wrong in the application. Recurring protective stops should be resolved, ignoring it can void warranty.

EXPLANATION

Frequent protective stops is an indication of either an error in the application, or a hardware failure in a joint.

SUGGESTION

Check that the motions of the robot do not cause collision, check payload, center of gravity and acceleration settings, if these settings are correct, the joint needs repair.

C163A5 Wrist 3 joint. Something is wrong in the application. Recurring protective stops should be resolved, ignoring it can void warranty.

EXPLANATION

Frequent protective stops is an indication of either an error in the application, or a hardware failure in a joint.

SUGGESTION

Check that the motions of the robot do not cause collision, check payload, center of gravity and acceleration settings, if these settings are correct, the joint needs repair.

1.83. C164 Wrist position close to safety plane limits

C164A0 Wrist 1 is too close to safety plane

C164A1 Wrist 2 is too close to safety plane

C164A2 Wrist 3 is too close to safety plane

1.84. C171 Issue with blends

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C171A0 A MoveC-Waypoint were skipped due to a blend.

EXPLANATION

The value for the blend radius is too large compared to the distance between the Waypoints.

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A1 Blend radius too small in a MoveC

SUGGESTION

Increase blend in MoveC

C171A3 A ServoC-Waypoint were skipped due to a blend.

EXPLANATION

The value for the blend radius is too large compared to the distance between the Waypoints.

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A4 Overlapping Blends in a MoveJ, a Waypoint was skipped

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A5 Overlapping Blends in a MoveJ, a Waypoint was skipped



SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A6 Overlapping Blends in a MoveJ, a Waypoint was skipped

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A7 Overlapping Blends in a MoveJ, a Waypoint was skipped

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A9 A MoveP-Waypoint were skipped due to a blend.

EXPLANATION

The value for the blend radius is too large compared to the distance between the Waypoints.

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A10 Blend radius too small error in a MoveP

C171A11 Overlapping Blends in a MoveL, a Waypoint was skipped

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A12 Overlapping Blends in a MoveL, a Waypoint was skipped

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A13 Overlapping Blends in a MoveL, a Waypoint was skipped

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

C171A14 Overlapping Blends in a MoveL, a Waypoint was skipped

SUGGESTION

Decrease the blend radius or choose Waypoints that are further apart.

1.85. C172 Illegal control mode

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.86. C173 Robot motion causes too high joint torques

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C173A0 Base.

C173A1 Shoulder.

C173A2 Elbow.

C173A3 Wrist 1.

C173A4 Wrist 2.

C173A5 Wrist 3.

C173A6 Base. Problem identified when executing program line {unsigned}.

C173A7 Shoulder. Problem identified when executing program line {unsigned}.

C173A8 Elbow. Problem identified when executing program line {unsigned}.
C173A9 Wrist 1. Problem identified when executing program line {unsigned}.
C173A10 Wrist 2. Problem identified when executing program line {unsigned}.
C173A11 Wrist 3. Problem identified when executing program line {unsigned}.

1.87. C174 Robot motion causes too high jump in joint torques

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C174A0 Base.
C174A1 Shoulder.
C174A2 Elbow.
C174A3 Wrist 1.
C174A4 Wrist 2.
C174A5 Wrist 3.
C174A6 Base. Problem identified when executing program line {unsigned}.
C174A7 Shoulder. Problem identified when executing program line {unsigned}.
C174A8 Elbow. Problem identified when executing program line {unsigned}.
C174A9 Wrist 1. Problem identified when executing program line {unsigned}.
C174A10 Wrist 2. Problem identified when executing program line {unsigned}.
C174A11 Wrist 3. Problem identified when executing program line {unsigned}.

1.88. C184 Joint self test not received by controller

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.89. C185

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C185A1 START_NORMAL_OPERATION is not allowed on selftest firmware
C185A2 GOTO_BACKDRIVE_COMMAND is not allowed on selftest firmware

1.90. C186

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C186A1 joint_mode == JOINT_RUNNING_MODE is not allowed on selftest firmware

1.91. C187 Temperature sensor test failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C187A1 Starting temperature were lower than expected



C187A2 Starting temperature were higher than expected

C187A3 Temperature increased less than expected during warm up

C187A4 Temperature increased more than expected during warm up

1.92. C190 Joint failed during selftest

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C190A0 Motor encoder index mark not found

C190A1 Phases not mounted correctly

C190A2 Motor encoder counting the wrong way

C190A3 Joint encoder counting the wrong way

C190A4 No movement detected while trying to move the motor

C190A11 Temperature alignment did not warm up to 45 degrees C within 30 minutes

C190A12 Temperature alignment did not cool down to 45 degrees C within 60 minutes

1.93. C191 Safety system violation

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C191A1 Joint position limit violated

C191A2 Joint speed limit violated

C191A3 TCP speed limit violated

C191A4 TCP position limit violated

C191A5 TCP orientation limit violated

C191A6 Power limit violated

C191A7 Joint torque window violated

C191A8 Joint torque window too large

C191A9 Reduced mode output violation

C191A10 Safeguard stop output violation

C191A11 Emergency stop output violation

C191A12 Momentum limit violation

C191A13 Robot moving output violation

C191A14 Robot is not braking in stop mode

EXPLANATION

During the braking process, the safety system monitors if the robot brakes as expected. If this is not the case, this error is generated

SUGGESTION

Check TCP configuration, payload, and mounting settings

C191A15 Robot is moving in stop mode

EXPLANATION

When the robot is stopped due to a safety violation or a safeguard stop, the safety system generates this error, if the robot moves while in this mode

SUGGESTION

(A) Check if the robot is physically pushed while safeguard stopped, (B) Check TCP configuration, payload, and mounting settings

C191A16 Robot did not stop in time

C191A17 Received a null vector for TCP orientation

C191A18 Robot not stopping output violation

C191A19 Invalid safety IO configuration

C191A20 Configuration information or limit sets not received

C191A21 The other safety processor detected a violation

C191A22 Received unknown command from Controller

C191A23 Invalid setup of safety limits

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check Firmware/update firmware, (B) Conduct a complete rebooting sequence

C191A24 Reduced Mode Output set, while it should not be

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check Firmware/update firmware, (B) Conduct a complete rebooting sequence

C191A25 Reduced Mode Output not set, while it should be

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check Firmware/update firmware, (B) Conduct a complete rebooting sequence

C191A26 Not Reduced Mode Output set, while it should not be

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check Firmware/update firmware, (B) Conduct a complete rebooting sequence

C191A27 Not Reduced Mode Output not set, while it should be

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check Firmware/update firmware, (B) Conduct a complete rebooting sequence

C191A28 Robot Emergency Stop exceeded maximum stop time

EXPLANATION

Too high payload

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check that max payload of the robot has not been exceeded, (B) Check TCP configuration, payload, and mounting settings

C191A29 System Emergency Stop exceeded maximum stop time

EXPLANATION

Too high payload

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Check that max payload of the robot has not been exceeded, (B) Check TCP configuration, payload, and mounting settings

C191A30 Safeguard Stop exceeded maximum stop time**EXPLANATION**

Too high payload

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check that max payload of the robot has not been exceeded, (B) Check TCP configuration, payload, and mounting settings

C191A31 Operation mode switch is present while the three position switch is missing**C191A32 Joint speed limit violated - Base****C191A33 Joint speed limit violated - Shoulder****C191A34 Joint speed limit violated - Elbow****C191A35 Joint speed limit violated - Wrist 1****C191A36 Joint speed limit violated - Wrist 2****C191A37 Joint speed limit violated - Wrist 3**

1.94. C192 Safety system fault

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C192A1 Robot still powered in emergency stop**EXPLANATION**

When the emergency stop is active, the Robot Arm powers off. The controller is responsible for sending the power off command. This error is generated if the safety system detects that the Robot Arm still has power

C192A2 Robot emergency stop disagreement**EXPLANATION**

E-stop in teach pendant or in robot E-stop circuit problem

SUGGESTION

Check safety devices and cables/connections to these devices.

C192A3 System emergency stop disagreement**EXPLANATION**

System E-stop circuit problem

SUGGESTION

Check safety devices and cables/connections to these devices.

C192A4 Safeguard stop disagreement**EXPLANATION**

Safeguard circuit problem

SUGGESTION

Check safety devices and cables/connections to these devices.

C192A5 Euromap safeguard stop disagreement**EXPLANATION**

Euromap circuit problem

SUGGESTION

Check cables from Safety Control Board to Euromap and to external machine

C192A6 Joint position disagreement

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check TCP configuration, payload and mounting settings, (B) Check that safety settings respected

C192A7 Joint speed disagreement

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check TCP configuration, payload and mounting settings, (B) Check that safety settings respected

C192A8 Joint torque disagreement

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check TCP configuration, payload and mounting settings, (B) Check that safety settings respected

C192A9 TCP speed disagreement

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check TCP configuration, payload and mounting settings, (B) Check that safety settings respected

C192A10 TCP position disagreement

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check TCP configuration, payload and mounting settings, (B) Check that safety settings respected

C192A11 TCP orientation disagreement

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check TCP configuration, payload and mounting settings, (B) Check that safety settings respected

C192A12 Power disagreement

EXPLANATION

Power calculation: uP-A and uP-B disagreement

C192A13 Joint torque window disagreement

C192A14 Reduced mode input disagreement

EXPLANATION

Safety I/O uP-A and uP-B disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A15 Reduced mode output disagreement

EXPLANATION

Safety I/O uP-A and uP-B disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A16 Safety output failed

EXPLANATION

The safety output did not reach the correct value in the expected time

SUGGESTION

Check for short circuit on I/O or for wrong connection to output.

C192A17 Safeguard stop output disagreement



EXPLANATION

Safety I/O uP-A and uP-B disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A18 The other safety processor is in fault

C192A19 Emergency stop output disagreement

EXPLANATION

Safety I/O uP-A and uP-B disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A20 SPI output error detected

EXPLANATION

Powersupply for the I/O is not detected

SUGGESTION

Check if the connection to the internal power supply is correct. If an external power supply is being used, check if it is powered on and at the correct voltage.

C192A21 Momentum disagreement

C192A22 Robot moving output disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A23 Wrong processor ID

C192A24 Wrong processor revision

C192A25 Potential brownout detected

EXPLANATION

Voltage drop on Safety Control Board or defect Safety Control Board

C192A26 Emergency stop output disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A27 Safeguard stop output disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A28 Robot not stopping output disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A29 Safeguard reset input disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A30 Safety processor booted up in fault mode

C192A31 Reduced Mode Output disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A32 Not Reduced Mode Output disagreement

SUGGESTION

Check safety devices and cables/connections to these devices

C192A33 A timing issue occurred during startup. Please restart to proceed**EXPLANATION**

Checksum disagreement between safety processors uA and uB

C192A34 User safety config checksum disagreement between uA and GUI**C192A35 Robot config checksum disagreement between uA and GUI****C192A36 Online RAM test failed****C192A37 Not all safety related functionalities are running****C192A38 Package too short for CRC calculation****C192A39 Three position switch input disagreement****C192A40 Operation mode switch input disagreement**

1.95. C193 One of the nodes is in fault mode

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C193A0 Base Joint**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Update the firmware on the joint, (C) Conduct a complete rebooting sequence

C193A1 Shoulder Joint**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Update the firmware on the joint, (C) Conduct a complete rebooting sequence

C193A2 Elbow Joint**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Update the firmware on the joint, (C) Conduct a complete rebooting sequence

C193A3 Wrist 1 Joint**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Update the firmware on the joint, (C) Conduct a complete rebooting sequence

C193A4 Wrist 2 Joint**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Update the firmware on the joint, (C) Conduct a complete rebooting sequence

C193A5 Wrist 3 Joint**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Update the firmware on the joint, (C) Conduct a complete rebooting sequence

C193A6 Tool

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Conduct a complete rebooting sequence

C193A7 Screen 1**EXPLANATION**

Safety Control Board has detected an error on Safety processor 1 in Teach pendant

SUGGESTION

Try the following actions to see which resolves the issue: (A) See previous error, (B) Conduct a complete rebooting sequence

C193A8 Screen 2**EXPLANATION**

Safety Control Board has detected an error on Safety processor 2 in Teach pendant

SUGGESTION

Try the following actions to see which resolves the issue: (A) See previous error, (B) Conduct a complete rebooting sequence

C193A9 Euromap 1**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Conduct a complete rebooting sequence

C193A10 Euromap 2**SUGGESTION**

Try the following actions to see which resolves the issue: (A) See previous error, (B) Conduct a complete rebooting sequence

1.96. C194 One of the nodes is not booted or not present

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C194A0 Base Joint**C194A1 Shoulder Joint****C194A2 Elbow Joint****C194A3 Wrist 1 Joint****C194A4 Wrist 2 Joint****C194A5 Wrist 3 Joint****C194A6 Tool****C194A7 Screen 1****EXPLANATION**

Safety Control Board has detected an error on Safety processor 1 in Teach pendant

C194A8 Screen 2**EXPLANATION**

Safety Control Board has detected an error on Safety processor 2 in Teach pendant

C194A9 Euromap 1**C194A10 Euromap 2****C194A128 Base not ready while brake release requested**

EXPLANATION

Must be at least in IDLE mode when the brake release is requested

SUGGESTION

Check for loose communication cable

C194A129 Shoulder not ready while brake release requested**EXPLANATION**

Must be at least in IDLE mode when the brake release is requested

SUGGESTION

Check for loose communication cable

C194A130 Elbow not ready while brake release requested**EXPLANATION**

Must be at least in IDLE mode when the brake release is requested

SUGGESTION

Check for loose communication cable

C194A131 Wrist 1 not ready while brake release requested**EXPLANATION**

Must be at least in IDLE mode when the brake release is requested

SUGGESTION

Check for loose communication cable

C194A132 Wrist 2 not ready while brake release requested**EXPLANATION**

Must be at least in IDLE mode when the brake release is requested

SUGGESTION

Check for loose communication cable

C194A133 Wrist 3 not ready while brake release requested**EXPLANATION**

Must be at least in IDLE mode when the brake release is requested

SUGGESTION

Check for loose communication cable

C194A134 Tool not ready while brake release requested**EXPLANATION**

Must be at least in IDLE mode when the brake release is requested

SUGGESTION

Check for loose communication cable

1.97. C195 Conveyor speed too high

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C195A1 for joint speed safety limit

C195A2 for TCP speed safety limit

C195A3 for momentum safety limit

1.98. C196 MoveP speed too high

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.



1.99. C197 Blend overlap warning

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.100. C200 Safety Control Board hardware error

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C200A1 Hardware ID is wrong

EXPLANATION

Safety Control Board: uP-A has detected an error: Wrong Safety Control Board

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A2 MCU type is wrong

EXPLANATION

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A3 Part ID is wrong

EXPLANATION

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A4 RAM test failed

EXPLANATION

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A5 Register test failed

EXPLANATION

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A6 pRom Crc test failed

EXPLANATION

Safety Control Board: uP-A has detected an error: firmware error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A7 Watchdog reset the processor**EXPLANATION**

Safety Control Board: uP-A has detected an error

C200A8 OVG signal test not passed**EXPLANATION**

Safety Control Board: uP-A has detected an error: over voltage generator

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A9 3V3A power good pin is low**EXPLANATION**

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A10 3V3B power good pin is low**EXPLANATION**

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A11 5V power good is low**EXPLANATION**

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A12 3V3 voltage too low**EXPLANATION**

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A13 3v3 voltage too high**EXPLANATION**

Safety Control Board: uP-A has detected an error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) If this happens more than twice in a row, replace Safety Control Board

C200A14 48V input is too low**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check 48 V Power supply, current distributor, energy eater and Safety Control Board for issues



C200A15 48V input is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check 48 V Power supply, current distributor, energy eater and Safety Control Board for issues

C200A16 24V IO short circuited

EXPLANATION

Too high current

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Disconnect external I/O connections and check external power supply if connected

C200A17 PC current is too high

EXPLANATION

Motherboard takes too high current

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check cable between Safety Control Board and Motherboard and check all connections to Motherboard. Also check for short circuit

C200A18 Robot voltage is too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for short circuit in Robot Arm, (C) Check 48 V Power supply, current distributor, energy eater and Safety Control Board for issues

C200A19 Robot voltage is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check 48 V Power supply, current distributor, energy eater and Safety Control Board for issues

C200A20 24V IO voltage is too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Disconnect I/O, check external power supply if connected and check Safety Control Board for issues

C200A21 12V voltage is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check 12 V Power supply, cables and Safety Control Board for issues

C200A22 12V voltage is too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check 12 V Power supply, cables and Safety Control Board for issues

C200A23 It took too long to stabilize 24V

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check 24 V and Safety Control Board for issues

C200A24 It took too long to stabilize 24V IO

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check 24 V and Safety Control Board for issues

C200A25 24V voltage is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check external 24 V and Safety Control Board for issues, (B) Conduct a complete rebooting sequence, (C) If this happens more than twice in a row, replace Safety Control Board.

C200A26 24V IO voltage is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disconnect I/O's, (B) Conduct a complete rebooting sequence, (C) Check external 24 V and Safety Control Board for issues

1.101. C201 Setup of Safety Control Board failed

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C201A0 Setup of Safety Control Board failed

EXPLANATION

No data was received from the Safety Control Board at initialization or invalid safety parameters have been received

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check that the ethernet cable between Motherboard and Safety Control Board is connected and verify that the setup of the Safety Configuration is valid.

C201A1 SCB uA is not responding

EXPLANATION

No data or invalid data was received from the Safety Control Board uA at initialization

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check that the ethernet cable between Motherboard and Safety Control Board is connected and verify that the setup of the Safety Configuration is valid.

C201A2 SCB uB is not responding

EXPLANATION

No data or invalid data was received from the Safety Control Board uB at initialization

SUGGESTION

Conduct a complete rebooting sequence.

C201A3 SCB is not responding

EXPLANATION

No data or invalid was received from Safety Control Board when requested for configuration parameters

SUGGESTION

Conduct a complete rebooting sequence.

1.102. C202 SCE configuration was illegal, after applying tolerances

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.



1.103. C203 PolyScope detected a mismatch between the shown and (to be) applied safety parameters

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.104. C204 Path sanity check failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C204A1 Sudden change in target position

C204A2 Inconsistency between target position and speed

C204A3 Sudden stop

EXPLANATION

The program contains motions that are not ramped correctly down

SUGGESTION

To abort a motion, use "stopj" or "stopl" script commands to generate a smooth deceleration before using "wait". Avoid aborting motions between Waypoints with blend

C204A4 Robot has not stopped in the allowed reaction and braking time

C204A5 Robot program resulted in invalid setpoint

C204A6 Blending failed and resulted in an invalid setpoint

SUGGESTION

Try the following actions to see which resolves the issue: (A) Try changing the blend radius, (B) Contact your local Universal Robots technical support

C204A7 Robot approaching singularity - Acceleration threshold failed

1.105. C205 Target speed does not match target position

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C205A0 Inconsistency between target position and speed

1.106. C206 Sanity check failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C206A0 Target joint speed does not match target joint position change - Base

C206A1 Target joint speed does not match target joint position change - Shoulder

C206A2 Target joint speed does not match target joint position change - Elbow

C206A3 Target joint speed does not match target joint position change - Wrist 1

C206A4 Target joint speed does not match target joint position change - Wrist 2

C206A5 Target joint speed does not match target joint position change - Wrist 3

1.107. C207 Fieldbus input disconnected

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.108. C208 Debug Assertion failed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.109. C209 A protective stop was triggered (for test purposes only)

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.110. C210 Socket is read-only when the robot is in local (Teach pendant) control

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.111. C211 Operational mode changed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C211A0 Disabled

C211A1 Automatic

C211A2 Manual

1.112. C212 Name conflict in loaded program

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C212A1 {unsigned} name conflict(s) occurred between feature names and program variables

EXPLANATION

Some feature names and program variables share the same name, which may cause confusion.

SUGGESTION

Rename the program variables.

1.113. C213 No Kinematic Calibration found (calibration.conf file is either corrupt or missing)

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.



1.114. C214 Kinematic Calibration for the robot does not match the joint(s)

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C214A1 The Kinematic Calibration checksum does not match the Base checksum

C214A2 The Kinematic Calibration checksum does not match the Shoulder checksum

C214A3 The Kinematic Calibration checksum does not match the Elbow checksum

C214A4 The Kinematic Calibration checksum does not match Wrist 1 checksum

C214A5 The Kinematic Calibration checksum does not match for Wrist 2 checksum

C214A6 The Kinematic Calibration checksum does not match for Wrist 3 checksum

1.115. C215 Kinematic Calibration does not match the robot

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.116. C216 The offset of the joint has changed

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C216A1 Base

C216A2 Shoulder

C216A3 Elbow

C216A4 Wrist 1

C216A5 Wrist 2

C216A6 Wrist 3

1.117. C217 White space detected at the beginning of a string token at line {unsigned}

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.118. C218 A thread used a lot of time

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C218A0 Main Robot Program.

C218A1 Thread: {string}

1.119. C219 Path Offset

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C219A1 Change in offset is too high to meet joint speed safety limit

EXPLANATION

Following the specified offsets would result in the robot exceeding safety limits

SUGGESTION

Reduce the change in offset between steps and/or apply more offset filtering using `path_offset_set_alpha_filter()`

C219A2 Change in offset is too high to meet tool speed safety limit

EXPLANATION

Following the specified offsets would result in the robot exceeding safety limits

SUGGESTION

Reduce the change in offset between steps and/or apply more offset filtering using `path_offset_set_alpha_filter()`

C219A3 Change in offset is too high to meet momentum safety limit

EXPLANATION

Following the specified offsets would result in the robot exceeding safety limits

SUGGESTION

Reduce the change in offset between steps and/or apply more offset filtering using `path_offset_set_alpha_filter()`

1.120. C220 Kinematic Calibration

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C220A1 Version {unsigned} on the robot arm is not supported

EXPLANATION

The kinematic calibration saved on the robot arm is an unsupported version.

SUGGESTION

(A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C220A2 Kinematic Calibration file was replaced with file from the arm.

EXPLANATION

The calibration.conf file was overwritten with the version stored in the arm. This is likely because the arm connected to the control box was changed

C220A3 Kinematic Calibration uploaded to the arm.

EXPLANATION

The calibration.conf file was changed and uploaded to the arm.

C220A4 Kinematic Calibration reuploaded to the arm.

EXPLANATION

The kinematic calibration was reuploaded to the arm as not all joints matching the calibration had it saved.

1.121. C221 GUI Communication

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C221A0 High load, messages dropped

EXPLANATION

GUI Communication is under high load causing user messages to be dropped

**SUGGESTION**

(A) Reduce the number of `textmsg()` and `varmsg()` being executed in one time-step (B) Insert `wait()` or `sync()` in the thread or main program

C221A1 Overload**EXPLANATION**

GUI Communication is overloaded

SUGGESTION

(A) Reduce the number of `textmsg()` and `varmsg()` being executed in one time-step (B) Insert `wait()` or `sync()` in the thread or main program

C221A2 A variable is too large to be sent to polyscope.**EXPLANATION**

The size of the variable is too big to be sent and cannot be shown in the variables tab or saved as an installation variable. This does not affect any other functionality.

SUGGESTION

To use the list as an installation variable or in the variable tab, make sure to limit its size by dividing it into smaller lists.

1.122. C222 Frame Tracking

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C222A1 Change in offset is too high to meet joint speed safety limit**EXPLANATION**

Tracking the specified frame would result in the robot exceeding safety limits

SUGGESTION

Smooth or slow the tracked frame if possible. Additionally, avoid singular arm configurations.

C222A2 Change in offset is too high to meet tool speed safety limit**EXPLANATION**

Tracking the specified frame would result in the robot exceeding safety limits

SUGGESTION

Smooth or slow the tracked frame if possible. Additionally, avoid singular arm configurations.

C222A3 Change in offset is too high to meet momentum safety limit**EXPLANATION**

Tracking the specified frame would result in the robot exceeding safety limits

SUGGESTION

Smooth or slow the tracked frame if possible. Additionally, avoid singular arm configurations.

C222A4 Change in offset is too high to meet joint speed safety limit**EXPLANATION**

Tracking the moving external axis would result in the robot exceeding safety limits

SUGGESTION

Reduce the acceleration of the external axis if possible. Additionally, avoid singular arm configurations.

C222A5 Change in offset is too high to meet tool speed safety limit**EXPLANATION**

Tracking the moving external axis would result in the robot exceeding safety limits

SUGGESTION

Reduce the acceleration of the external axis if possible. Additionally, avoid singular arm configurations.

C222A6 Change in offset is too high to meet momentum safety limit

EXPLANATION

Tracking the moving external axis would result in the robot exceeding safety limits

SUGGESTION

Reduce the acceleration of the external axis if possible. Additionally, avoid singular arm configurations.

1.123. C223 Flexible EtherNet/IP Fieldbus input (custom instance) disconnected

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.124. C224 {string}

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

1.125. C225 Logic Program '{string}' is not in running state.

1.126. C226 A fault was triggered (for test purposes only)



2. Error Codes e-Series and Later Models

Error codes with numbers above 256 apply only to e-Series and later models.

2.1. C259 Filesystem-related issue

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C259A0 Critical error

EXPLANATION

A critical error occurred in the filesystem

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Reimage SD card, (C) Contact your local Universal Robots service provider for assistance

C259A80

EXPLANATION

The data stored in joint is obsolete

SUGGESTION

Updating firmware needs to be done incrementally from such old persistent joint data. Re-deploy the previous firmware, and perform incremental upgrade up to SW 5.5 before deploying this version. **IMPORTANT:** Power on the robot arm between each update and validate it reaches IDLE state. Contact your local Universal Robots service provider for assistance

C259A89 Failed to create a file with ID {unsigned}

EXPLANATION

The joint has failed to create a file in its internal filesystem

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.2. C260 Brake Release – old procedure

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C260A0 Critical error

EXPLANATION

A critical error occurred during Brake Release

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Ensure the mounted payload, TCP, and CoG matches your configuration, (C) Contact your local Universal Robots service provider for assistance

2.3. C261 Temperature Sensor

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C261A0 Critical error

EXPLANATION

A critical error occurred in the temperature sensor

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence (B) Contact your local Universal Robots service provider for assistance

C261A3 Temperature changed more than allowed: {float} Celsius

EXPLANATION

The maximum allowed difference between two sequential temperature readings was exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence (B) Update software (C) Contact your local Universal Robots service provider for assistance

C261A4 Temperature is too high ({float} degrees Celsius)

EXPLANATION

Ambient temperature is too high or robot is overloaded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Cool down the robot's environment and make sure the robot is operating within recommended limits, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C261A5 Temperature is too low ({float} degrees Celsius)

EXPLANATION

Ambient temperature is too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the robot is operating within recommended limits, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C261A21 Safety Control Board temperature is too high: {float} Celsius

EXPLANATION

The temperature of the Safety Control Board is too high.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Safety Control Board (E) Contact your local Universal Robots service provider for assistance

C261A22 Safety Control Board temperature is too low: {float} Celsius

EXPLANATION

The temperature of the Safety Control Board is too low.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Increase ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Safety Control Board (E) Contact your local Universal Robots service provider for assistance



C261A23 The Energy Removal power sensor temperature on the Power Board is too high: {float} Celsius

EXPLANATION

The temperature of the Energy Removal power sensor on the Power Board is too high.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce excessive decelerations in the robot program, (B) Reduce ambient temperature (C) Conduct a complete rebooting sequence, (D) Update software, (E) Replace the Power Board (F) Contact your local Universal Robots service provider for assistance

C261A24 The Energy Removal power sensor temperature on the Power Board is too low: {float} Celsius

EXPLANATION

The temperature of the Energy Removal power sensor on the Power Board is too low.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Increase ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Power Board (E) Contact your local Universal Robots service provider for assistance

C261A25 The Robot power sensor temperature on the Power Board is too high: {float} Celsius

EXPLANATION

The temperature of the Robot power sensor on the Power Board is too high.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce excessive decelerations in the robot program, (B) Reduce ambient temperature (C) Conduct a complete rebooting sequence, (D) Update software, (E) Replace the Power Board (F) Contact your local Universal Robots service provider for assistance

C261A26 The Robot power sensor temperature on the Power Board is too low: {float} Celsius

EXPLANATION

The temperature of the Robot power sensor on the Power Board is too low.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Increase ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Power Board (E) Contact your local Universal Robots service provider for assistance

C261A29 Unable to read the TOOL PCBA temperature for {float} ms.

EXPLANATION

The TOOL node has been unable to read the PCBA temperature sensor.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Replace the TOOL PCBA (C) Contact your local Universal Robots service provider for assistance

C261A30 Safety Control Board temperature changed more than allowed: {float} Celsius

EXPLANATION

The maximum allowed difference between two sequential temperature readings was exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence (B) Update software (C) Contact your local Universal Robots service provider for assistance

C261A31 Energy Removal power sensor temperature on the Power Board changed more than allowed: {float} Celsius

EXPLANATION

The maximum allowed difference between two sequential temperature readings was exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence (B) Update software (C) Contact your local Universal Robots service provider for assistance

C261A32 PSU 48V circuit temperature is too high: {float} Celsius**EXPLANATION**

The temperature of the 48V circuit in the Power Supply Unit is too high.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Power Supply Unit (E) Contact your local Universal Robots service provider for assistance

C261A33 PSU 48V circuit temperature is too low: {float} Celsius**EXPLANATION**

The temperature of the 48V circuit in the Power Supply Unit is too low.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Power Supply Unit (E) Contact your local Universal Robots service provider for assistance

C261A34 PSU 48V circuit temperature changed more than allowed: {float} Celsius**EXPLANATION**

The maximum allowed difference between two sequential temperature readings was exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence (B) Update software (C) Contact your local Universal Robots service provider for assistance

C261A35 PSU 24V circuit temperature is too high: {float} Celsius**EXPLANATION**

The temperature of the 24V circuit in the Power Supply Unit is too high.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Power Supply Unit (E) Contact your local Universal Robots service provider for assistance

C261A36 PSU 24V circuit temperature is too low: {float} Celsius**EXPLANATION**

The temperature of the 24V circuit in the Power Supply Unit is too low.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Power Supply Unit (E) Contact your local Universal Robots service provider for assistance

C261A37 PSU 24V circuit temperature changed more than allowed: {float} Celsius**EXPLANATION**

The maximum allowed difference between two sequential temperature readings was exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence (B) Update software (C) Contact your local Universal Robots service provider for assistance

C261A38 PSU PFC circuit temperature is too low: {float} Celsius**EXPLANATION**

The temperature of the PFC circuit in the Power Supply Unit is too high.

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Reduce ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Power Supply Unit (E) Contact your local Universal Robots service provider for assistance

C261A39 PSU PFC circuit temperature is too low: {float} Celsius**EXPLANATION**

The temperature of the PFC circuit in the Power Supply Unit is too low.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce ambient temperature (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace the Power Supply Unit (E) Contact your local Universal Robots service provider for assistance

C261A40 PSU PFC circuit temperature changed more than allowed: {float} Celsius**EXPLANATION**

The maximum allowed difference between two sequential temperature readings was exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence (B) Update software (C) Contact your local Universal Robots service provider for assistance

2.4. C262 Communication

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C262A0 Critical error**EXPLANATION**

A critical error occurred in the communication framework

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for ESD noise, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C262A17 Failed to communicate with {deviceName} Joint**EXPLANATION**

Joint failed to start communicating as expected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the cable between Control Box and Robot Arm is correctly connected and it has no damage, (B) Check for loose connections, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C262A18 Failed to communicate with TOOL**EXPLANATION**

Tool failed to start communicating as expected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C262A25 Unexpected message version received: {unsigned}**EXPLANATION**

The received message version is not supported.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance.

C262A27 Failed to communicate with the Base Filter Board**EXPLANATION**

The Base Filter Board failed to start communicating as expected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C262A28 Failed to properly instantiate a multi-subscriber message or special command**EXPLANATION**

The value use to store the element must be placed at the persistence RAM, i.e. not on the stack

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C262A29 Failed to communicate with the Cable Extender with ID: {hex}**EXPLANATION**

The Cable Extender Board failed to start communicating as expected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.5. C263 Motor Encoder

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C263A0 Critical error**EXPLANATION**

A critical error occurred in the Motor Encoder.

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C263A1 Motor Encoder is unavailable**EXPLANATION**

The Motor Encoder's sense signal indicates a bad connection.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C263A2 Calibration has been invalidated and can lead to reduced performance.**EXPLANATION**

An error in the file handling caused the invalid calibration, which can alter joint performance.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C263A21 Validation of the detected Index Mark ({signed}) failed

**EXPLANATION**

The absolute Motor Encoder position cannot be validated because the Motor position calculated by the Joint Encoder is invalid.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C263A26 Failure to log missing Index Mark, index out of range: {unsigned}**EXPLANATION**

Failed to log the missing index mark position because the index is out of range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C263A27 Failure to log index drift, position out of range: {unsigned}**EXPLANATION**

Failed to log the index drift position because the position is out of range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C263A38 Time sanity check failed, difference is too great: {float}**EXPLANATION**

The time difference (in μ s) since the last motor encoder read is too great

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C263A39 The motor encoder data CRC was invalid**EXPLANATION**

The CRC check of the motor encoder data failed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C263A41 The rate of invalid samples is too high. Debt counter = {unsigned}**EXPLANATION**

A high rate of invalid samples can indicate a faulty motor encoder, a bad connection, or an EMC/ESD issue.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for possible EM noise or ESD sources, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

2.6. C264 Task Manager

You may find the following links helpful in addressing this issue:

- [How to perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C264A0 Critical error

EXPLANATION

A critical error occurred in the task manager

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C264A7 Start of Cycle pulse was required but did not occur after {unsigned}ms.**EXPLANATION**

A Start of Cycle pulse did not happen within the required time. The issue may be due to a damaged internal communication cable to the node or a bad internal connection.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C264A8 Systick timer and Start of Cycle pulse was misaligned by {float}us.**EXPLANATION**

The internal systick timer and the Start of Cycle pulse was not aligned as expected. The issue may be due to a damaged internal communication cable to the node or a bad internal connection.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C264A9 {signed} unexpected (+) or missing (-) Start of Cycle pulses was detected within a 1 second period.**EXPLANATION**

The number of Start of Cycle pulses within a 1 second period was different than expected. The issue may be due to a damaged internal communication cable to the node or a bad internal connection.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C264A10 {unsigned} invalid Start of Cycle pulses was detected within a 1 second period.**EXPLANATION**

Too many misaligned Start of Cycle pulses was detected within a 1 second period. The issue may be due to a damaged internal communication cable to the node or a bad internal connection.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C264A11 {unsigned} Start of Cycle pulses was lost in a row**EXPLANATION**

Too many Start of Cycle pulses was lost in a row. The issue may be due to a damaged internal communication cable to the node or a bad internal connection.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance



2.7. C265 Joint Encoder

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C265A0 Joint encoder position invalid. Detailed error: {hex}

EXPLANATION

A critical error occurred in the Joint Encoder. The reported position is not valid.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for ESD noise, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C265A5 Near operation limits. Status: {hex}

EXPLANATION

The joint encoder is close to operational limits. Reported positions have reduced precision.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for ESD noise, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C265A6 Not present. Status: {hex}

EXPLANATION

The Joint Encoder is not responding to commands

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for ESD noise, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C265A41 The read head temperature is outside the allowed range.

EXPLANATION

The encoder read head is too cold or hot, either the ambient temperature is too high or the robot is overloaded.

SUGGESTION

Cool down the environment and make sure the robot is operating within the recommended limits.

C265A42 Signal amplitude low.

EXPLANATION

The distance between the readhead and the ring is too large.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A43 Signal amplitude too high.

EXPLANATION

The readhead is too close to the ring.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A44 Signal decoding below confidence threshold - position decoding might be inaccurate or fail.

EXPLANATION

At least one-out-of-two decoding algorithms, from the sensor image, falls below the confidence threshold. The read head might be tilted.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A45 Internal speed data is not valid.**EXPLANATION**

The time delta between last two position requests exceeded the maximum time allowed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A46 Encoder acceleration too high.**EXPLANATION**

The position data changed unexpectedly. A stray magnetic field is present or metal particles are present between the read head and the ring.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Remove any sources of magnetic fields external to the robot, and conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A47 Magnetic pattern decoding error**EXPLANATION**

A stray magnetic field is present or metal particles are present between the read head and the ring or radial positioning between the read head and the ring is out of tolerances.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Remove any sources of magnetic fields external to the robot, and conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A48 Signal lost.**EXPLANATION**

Signal lost. The read head is out of alignment with the ring or the ring is damaged.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A49 Signal amplitude too high. External magnetic field is present**EXPLANATION**

External magnetic field disturbing the jointencoder position reading

SUGGESTION

Try the following actions to see which resolves the issue: (A) Remove any sources of magnetic fields external to the robot, and conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A50 System error. Malfunction inside the circuitry.**EXPLANATION**

Malfunction inside the circuitry. To reset the System error bit try to cycle the power supply. The power-supply rise time should be shorter than 20 ms.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance



C265A51 Power supply voltage out of range.

EXPLANATION

Internal power supply error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A52 System error. Inconsistent calibration data is detected.

EXPLANATION

Inconsistent calibration data is detected. To reset the System error bit try to cycle the power supply. The power-supply rise time should be shorter than 20 ms.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A53 Too many warnings in a row.

EXPLANATION

The joint encoder reporting too many warnings in row

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C265A58 Invalid sample rate too high.

EXPLANATION

The joint encoder produced too many invalid samples.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.8. C266 Self-test

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C266A0 Critical error

EXPLANATION

A critical error occurred in the Joint Self-test

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.9. C267 Bootloader error

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C267A0 Critical error

EXPLANATION

A critical error occurred during Firmware upgrade.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C267A11 Hardware configuration issue**EXPLANATION**

Hardware configuration mismatch

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C267A13 Required firmware file is missing for device ID: {hex}**EXPLANATION**

A firmware file for a device is missing and the device does not have the required information to boot

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C267A14 Device not supported by the firmware file, device ID: {hex}**EXPLANATION**

A firmware file does not support a device and the device does not have the required information to boot

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C267A15 No firmware to boot from for device ID: {hex}**EXPLANATION**

The device does not have the required information to boot

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.10. C268 Special Command

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C268A4 Reboot command received but the device is not allowed to reboot**EXPLANATION**

Device received a reboot request but the system is in a state where the device is not allowed to reboot

SUGGESTION

Contact your local Universal Robots service provider for assistance.

2.11. C269 Transceiver – deprecated

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C269A70 Flash device is not supported, JEDEC data for device is: {hex}

**EXPLANATION**

The flash device's JEDEC ID does not match a supported flash device

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.12. C271 Low level real-time thread

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C271A1 Runtime is too much behind.

EXPLANATION

One of the threads might be using too much time (see log for more details).

SUGGESTION

Consider using a Wait or sync() to split long program sequences that do not move the robot.

2.13. C272 Missing calibration

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C272A0 Critical error - the calibration is missing

EXPLANATION

Calibration could not be loaded

SUGGESTION

Contact your local Universal Robots service provider for assistance.

2.14. C273 Cross monitoring

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C273A0 Critical error

EXPLANATION

A critical disagreement error occurred in the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A5 Disagreement on Safety Control Board State

EXPLANATION

A critical disagreement occurred within the safety system.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A6 Disagreement on Robot State

EXPLANATION

A critical disagreement occurred within the safety system.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A7 Disagreement on Safety State**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A8 Disagreement on position**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A9 Disagreement on velocity**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A10 Disagreement on current**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A11 Disagreement on temperature**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A12 Disagreement on Teach Pendant State**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A13 Disagreement on Teach Pendant Emergency Stop**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A14 One processor entered Fault State

**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A15 One processor entered Violation State**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A16 Joint State disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A17 Joint Constant Data CRC disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A18 Joint target current disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A19 Torque Window disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A20 Torque Error disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A21 Target Velocity disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A22 Target Acceleration disagreement

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A23 Recovery Mode CRC disagreement

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A24 Robot Configuration CRC disagreement

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A25 User Configuration CRC disagreement

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A26 Maximum stopping time disagreement

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A27 Stopping Time Torque Overload disagreement

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A28 Disagreement error on joint {unsigned}

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A29 Tool speed disagreement

EXPLANATION

A critical disagreement occurred within the safety system

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A30 Safety Mode Limit disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A31 Hand Protection Distance disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A32 Elbow Sphere speed disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A33 Momentum disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A34 Power disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A35 Elbow position disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A36 Workpiece Rotation disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A37 Disagreement on Workpiece Position

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A38 Disagreement on motor parameter (R_pp)**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A39 Disagreement on motor parameter (L_pp)**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A40 Disagreement on motor parameter (Kb)**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A41 Disagreement on motor parameter (Kt)**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A42 Disagreement on motor parameter (T)**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A43 Disagreement on the Teach Pendant's Three-Position Enabling Device**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A44 Disagreement on the active status of the Teach Pendant's Three-Position Enabling Device**EXPLANATION**

A critical disagreement occurred within the safety system

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A47 Disagreement on state**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A48 Disagreement on Injection-Molding-Machine-Interface Emergency Stop input**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A49 Disagreement on Injection-Molding-Machine-Interface Emergency Stop output**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A50 Disagreement on Injection-Molding-Machine-Interface Safeguard input**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A51 Disagreement on Injection-Molding-Machine-Interface type**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A52 Disagreement on Torque Parameters CRC**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A53 Target Torque disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A54 Disagreement on hardware configuration CRC

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A55 Disagreement on compensation current**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A56 Disagreement on external torque target**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A57 Safety Target Torque disagreement**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A58 Disagreement on all motors off in arm**EXPLANATION**

A critical disagreement occurred within the safety system.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A61 Disagreement on joint gear temperature**EXPLANATION**

Safety system disagrees on the gear temperature

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A62 Disagreement on joint house temperature**EXPLANATION**

Safety system disagrees on the house temperature

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A63 Disagreement on the tool safety input**EXPLANATION**

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance



C273A64 One processor entered Critical Fault State

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A66 Disagreement on linear velocity

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR service provider for assistance

C273A67 Disagreement on angular velocity

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR service provider for assistance

C273A68 Disagreement on scanner monitoring case for scanner {unsigned}

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR service provider for assistance

C273A69 Disagreement on raw current

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C273A70 Safety field configuration CRC disagreement

EXPLANATION

A critical disagreement occurred within the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR service provider for assistance

2.15. C274 Control box fan error

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C274A1 Fan is not running

EXPLANATION

The control box fan must be running to prevent the control box from overheating

SUGGESTION

Ensure the fan in the control box can rotate freely and the connector is attached. The control box fan is located between the air filter and the energy eater in the control box. See the Service Manual for details.

C274A2 Monitoring data timed out**EXPLANATION**

The monitoring signals for the fan speed was not received by the system in a timely manner

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.16. C275 Brake Pin

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C275A12 Attempted to update brakepin state machine before it was initialized.**EXPLANATION**

Problem in joint firmware initialization sequence.

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.17. C276 Uart

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C276A0 Critical error**EXPLANATION**

A critical error occurred in the UART driver

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.18. C277 Memory

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C277A1 Failed to allocate memory**EXPLANATION**

Failed to allocate mamory.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance



2.19. C278 Servo

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C278A0 Critical error

EXPLANATION

A critical error occurred in the servo module

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C278A18 The version of the received control message is not supported. Version received: {unsigned}

EXPLANATION

The servo module received a control message from the controller of a version that it does not support

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C278A20 The control-mode state {unsigned} in the received control message is not supported.

EXPLANATION

The servo module received a control message from the controller with a control-mode state that it does not support

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.20. C279 Flash

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C279A0 Critical error

EXPLANATION

A critical error occurred in the flash driver

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C279A38 Timed out waiting for system voltage to reach {float}V

EXPLANATION

System voltage is too low to ensure safe flash operations

SUGGESTION

Try the following actions to see which resolves the issue: (A) Investigate power supply to the robot, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

2.21. C280 Real-time error

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C280A0 Critical error

EXPLANATION

A critical real-time error occurred

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Check communication between joints, (D) Contact your local Universal Robots service provider for assistance

2.22. C281 Robot State Machine

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C281A0 Critical error

EXPLANATION

A critical State Machine error occurred

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A3 {deviceName} joint entered the Fault State

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A4 {deviceName} joint entered the Violation State

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A7 Teach Pendant entered the Fault State

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A8 Teach Pendant entered the Violation State

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A9 {deviceName} joint moved too far before robot entered RUNNING State



EXPLANATION

A Joint moved more than the permissible range during the Brake Release procedure.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Ensure the mounted payload, TCP, and CoG matches your configuration, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C281A14 IMMI entered the Fault State

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A15 IMMI entered the Violation State

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A16 {deviceName} joint did not reach correct state before timeout was exceeded

EXPLANATION

The robot is not in the expected state. An issue occurred during the state transition.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C281A17 {deviceName} joint entered the Brake Failure State

EXPLANATION

The joint brake system failed to engage while powering down the robot.

SUGGESTION

Keep the robot powered on and contact your local Universal Robots service provider for assistance

C281A18 Not all joints reached parking within the expected time of {unsigned} ms

EXPLANATION

One or more joints did not transition to parking within the time allowed from when the first joint began to park

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A20 {deviceName} joint entered the Critical Fault State

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C281A21 Tool entered the Fault State

EXPLANATION

The tool has entered the Fault State

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.23. C283 Safety system

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C283A0 Critical error

EXPLANATION

A critical error occurred in the safety system

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A1 Robot is not braking when in Stop Mode

C283A2 Robot is moving when in Stop Mode

C283A3 Power not removed from the motors while in Emergency Stop

C283A4 Failed to power on the Robot Arm

C283A5 Invalid pin-configuration received: {hex}

C283A6 Trying to reassign pin configuration with configuration {hex}

C283A7 {deviceName} joint exceeded the speed limit of the safety settings

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A8 The System Emergency Stop Output is not active

EXPLANATION

Failed to activate the System Emergency Stop Output. The output is active when voltage is low

SUGGESTION

Make sure output is not short circuited to a power supply

C283A9 System Emergency Stop Output disagreement within the safety system

EXPLANATION

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A10 Robot Emergency Stop Input disagreement within the safety system

EXPLANATION

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A11 System Emergency Stop Input disagreement within the safety system

EXPLANATION

The input signals are not switching simultaneously, or are incorrectly connected.

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A12 Safeguard Stop Input disagreement within the safety system**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A13 Safeguard Reset Input disagreement within the safety system**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A14 Operation Mode input disagreement within the safety system.**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A15 Three-Positional Enabling Device Input disagreement within the safety system**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A16 Operation Mode Switch is defined and no Three-Positional Device is defined.**EXPLANATION**

Operation mode switch can only be enabled if a Three-Positional Device is present

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure that a Three-Positional Device is enabled in the Safety Configuration (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A17 Lost {unsigned} Teach Pendant safety packages in a row**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Check that the teach pendant is properly connected, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A18 Lost too many Joint safety packages in a row. Diagnostic data: {unsigned}**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Check that the teach pendant is properly connected, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A19 Invalid gravity vector received

EXPLANATION

The gravity vector is outside the allowed range.

SUGGESTION

Make sure there are no script commands that set an invalid gravity vector.

C283A20 Invalid payload mass received**EXPLANATION**

The payload mass is outside the allowed range

SUGGESTION

Make sure there are no script commands that set an illegal payload for the robot type

C283A21 invalid payload center of gravity received**EXPLANATION**

The payload center of gravity is outside the allowed range

SUGGESTION

Make sure there are no script commands that set an invalid payload center of gravity.

C283A22 Teach Pendant is connected while it is disabled in robot configuration**EXPLANATION**

If the Teach Pendant is enabled, it is connected. If it is disabled, it is not connected.

SUGGESTION

Disconnect the Teach Pendant or enable it in the configuration.

C283A26 Force limitation: A joint exceeded the torque window by {float}Nm**EXPLANATION**

If the target trajectory is very jerky, e.g. in case of sensor based control with `path_offset()` or `servoj()`, the target joint torques can exceed safety system limits

SUGGESTION

Check that the target trajectory is smooth. Noisy sensors or communication jitter in external control can cause jerky trajectories

C283A27 Mismatch on Robot Configuration CRC between the safety system and PolyScope**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A28 Mismatch on User Configuration CRC between the safety system and PolyScope**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Make new installation, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A34 Error while trying to apply safety configuration**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A35 Reduced Mode Output disagreement within the safety system**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A36 Not Reduced Mode Output disagreement within the safety system

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A37 Robot Moving Output disagreement within the safety system**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A38 Robot Not Stopping Output disagreement within the safety system**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A40 Reduced Mode Input disagreement within the safety systems**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are correctly connected, (B) Contact your local Universal Robots service provider for assistance.

C283A41 TCP Velocity violates limits of maximum stopping time**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A42 TCP Velocity violates limits of maximum stopping distance**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A43 {deviceName} joint moved too quickly toward a Joint position limit**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A44 The tool moved too fast towards an orientation limit**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A45 The Elbow moved too fast towards a safety plane**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A46 The tool moved too fast towards a safety plane**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A47 {deviceName} joint position limit exceeded

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A48 Tool position limit exceeded

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A49 Tool orientation limit exceeded

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A50 Elbow position limit exceeded

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A51 Robot moved with a speed of {float} mm/s at the tool. This exceeds the tool speed limit in the safety settings

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A52 Robot moved with a speed of {float} mm/s at the elbow. This exceeds the elbow speed limit in the safety settings

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A53 Maximum Tool Center Point Speed in Reduced Mode is invalid

SUGGESTION

Ensure the Reduced Mode Limit is less than or equal to the Normal Mode limit.

C283A54 Maximum Elbow Speed in Reduced Mode is invalid

SUGGESTION

Ensure the Reduced Mode limit is less than or equal to the Normal Mode Limit.

C283A55 Maximum Joint Speed of joint {unsigned} in Reduced Mode is invalid

SUGGESTION

Ensure the Reduced Mode Limit is less than or equal to the Normal Mode Limit.

C283A56 Maximum Momentum in Reduced Mode is invalid

SUGGESTION

Ensure the Reduced Mode Limit is less than or equal to the Normal Mode Limit.

C283A57 Maximum stopping time in Reduced Mode is invalid

**SUGGESTION**

Ensure the Reduced Mode Limit is less than or equal to the Normal Mode Limit.

C283A58 Maximum stopping distance in Reduced Mode is invalid**SUGGESTION**

Ensure the Reduced Mode Limit is less than or equal to the Normal Mode Limit.

C283A59 Reduced Mode Output is not active**EXPLANATION**

Failed to activate the Reduced Mode Output. The output is active when voltage is low

SUGGESTION

Make sure output is not short circuited to a power supply

C283A60 Reduced Mode Output is not inactive**EXPLANATION**

Failed to deactivate the Reduced Mode Output. The output is inactive when voltage is high

SUGGESTION

Make sure output is not short circuited to ground

C283A61 Not Reduced Mode Output is not active**EXPLANATION**

Failed to activate the Not Reduced Mode Output. The output is active when voltage is low

SUGGESTION

Make sure output is not short circuited to a power supply

C283A62 Not Reduced Mode Output is not inactive**EXPLANATION**

Failed to deactivate the Not Reduced Mode Output. The output is inactive when voltage is high

SUGGESTION

Make sure output is not short circuited to ground

C283A63 Robot is moving while Robot Moving Output is not active**EXPLANATION**

Failed to activate the Robot Moving Output. The output is active when voltage is low

SUGGESTION

Make sure output is not short circuited to a power supply

C283A64 Tool Direction Vector Length for Normal Mode is {float}, not 1.0**SUGGESTION**

Reconfigure Tool orientation

C283A65 Tool Direction Vector Length for Reduced Mode is {float}, not 1.0**SUGGESTION**

Reconfigure Tool orientation

C283A66 Robot Momentum reached {float} kg * m/s, which exceeds the Momentum limit**C283A67 Robot Power reached {float} W, which exceeds the Power limit****C283A68 Error caused by {deviceName}****EXPLANATION**

A critical safety error

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A72 The motor configuration sent by the Control Box is invalid**EXPLANATION**

The motor configuration sent is unusable with this firmware revision.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A73 Safe Home Position Output disagreement within the safety system**C283A74 The Safe Home Position Output is active while not allowed****EXPLANATION**

The Safe Home Position Output is active while the robot is not in Safe Home Position

SUGGESTION

Make sure output is not short circuited to power supply or ground

C283A81 The robot configuration specifies an unsupported joint size {signed}**EXPLANATION**

The safety system is not certified to work with the specified joint size

SUGGESTION

Upgrade to a newer software version

C283A82 The connected Teach Pendant type does not match the configuration**EXPLANATION**

The connected Teach Pendant is not the same type as the one selected in the safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check that the Teach Pendant is properly connected and matches the one in the safety configuration, (B) Conduct a complete rebooting sequence, (C) Update the software, (D) Contact your local Universal Robots technical support

C283A83 The configured Teach Pendant has no Three-Positional Enabling Device**EXPLANATION**

The safety configuration enables the Teach Pendant's Three-Positional Enabling Device, but the configured Teach Pendant does not have a Three-Positional Enabling Device

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check that the correct Teach Pendant is selected in the safety configuration, (B) Conduct a complete rebooting sequence, (C) Update the software, (D) Contact your local Universal Robots technical support

C283A85 Automatic Safeguard Stop Input disagreement within the safety system**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A86 Automatic Safeguard Reset Input disagreement within the safety system**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A87 Injection-Molding-Machine-Interface is connected while it is disabled in the robot configuration**EXPLANATION**

If the Injection-Molding-Machine-Interface is enabled, it must be connected. If it is disabled, it must be disconnected.

SUGGESTION

Disconnect the Injection-Molding-Machine-Interface or enable it in the configuration.

C283A88 Lost {unsigned} Injection-Molding-Machine-Interface safety packages in a row**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Check that the Injection-Molding-Machine-Interface is properly connected, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A89 The connected Injection-Molding-Machine-Interface type does not match the configuration**EXPLANATION**

The connected Injection-Molding-Machine-Interface is not the same type as the one selected in the safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check that the Injection-Molding-Machine-Interface is properly connected and matches the one in the safety configuration, (B) Conduct a complete rebooting sequence, (C) Update the software, (D) Contact your local Universal Robots technical support

C283A90 Invalid Injection-Molding-Machine-Interface type in the user configuration: {unsigned}**EXPLANATION**

The configuration provided by the user safety configuration is invalid

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure the correct IMMI type is selected in the Safety Configuration (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A91 The Injection-Molding-Machine-Interface System Emergency Stop Output is not active**EXPLANATION**

Failed to activate the System Emergency Stop Output on the Injection-Molding-Machine-Interface. The output is active when voltage is high

SUGGESTION

Make sure output is not short circuited to ground

C283A94 Automatic Safeguard Stop input is configured but no Three-Position Enabling device is configured**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Ensure that a Three-Positional Device is enabled in the Safety Configuration (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A97 The payload inertia matrix diagonal sent from the controller must be non-negative**SUGGESTION**

Make sure there are no script commands that set an invalid payload inertia matrix diagonal

C283A98 The payload inertia sent from the controller must be within valid range

EXPLANATION

The payload inertia must be within the valid range. Max inertia is defined as a 10 meter rod with a mass equal to the maximum rated robot payload

SUGGESTION

Make sure your inertia is within the valid range

C283A99 Received an invalid value {float} as part of the runtime safety configuration**SUGGESTION**

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A100 Multiple sources defined for controlling operational mode**SUGGESTION**

Try the following actions to resolve the issue: (A) Make sure there are not set multiple sources in the Safety Config controlling the operational mode, (B) Update Software, (C) Contact your local Universal Robots service provider for assistance

C283A102 The reduced mode state is inactive while not allowed**EXPLANATION**

The reduced mode state sent to the safety system is inactive while the robot is in reduced mode

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A103 The reduced mode state is active while not allowed**EXPLANATION**

The reduced mode state sent to the safety system is active while the robot is not in reduced mode

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A104 Robot is moving while the robot moving state is not active**EXPLANATION**

The robot is moving while the robot moving state sent to the safety system is inactive.

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A105 The safe home position state is active while not allowed**EXPLANATION**

The safe home position state sent to the safety system is active while the robot is not in safe home position

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A106 The safeguard stop state is active while not allowed**EXPLANATION**

The safeguard stop state sent to the safety system is active while the robot is not in safeguard stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A107 The safeguard stop state is inactive while not allowed**EXPLANATION**

The safeguard stop state sent to the safety system is inactive while the robot is in safeguard stop

**SUGGESTION**

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A112 The emergency stop by system state is active while not allowed**EXPLANATION**

The emergency stop by system state sent to the safety system is active while the robot is not in system emergency stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A113 The emergency stop by system state is inactive while not allowed**EXPLANATION**

The emergency stop by system state sent to the safety system is inactive while the robot is in system emergency stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A114 The emergency stop by robot state is active while not allowed**EXPLANATION**

The emergency stop by robot state sent to the safety system is active while the robot is not in robot emergency stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A115 The emergency stop by robot state is inactive while not allowed**EXPLANATION**

The emergency stop by robot state sent to the safety system is inactive while the robot is in robot emergency stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A116 The Fault state is inactive while not allowed**EXPLANATION**

The fault state sent to the safety system is inactive while the robot is in fault

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A117 The fault state is active while not allowed**EXPLANATION**

The fault state sent to the safety system is active while the robot is not in Fault

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A118 The violation state is inactive while not allowed**EXPLANATION**

The violation state sent to the safety system is inactive while the robot is in violation

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A119 The violation state is active while not allowed

EXPLANATION

The Violation state sent to the safety system is active while the robot is not in violation

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A120 The category 1 stop state is inactive while not allowed**EXPLANATION**

The category 1 stop state sent to the safety system is inactive while the robot is in a category 1 stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A121 The category 1 stop state is active while not allowed**EXPLANATION**

The category 1 stop state sent to the safety system is active while the robot is not in a category 1 stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A122 The safeguard stop auto state is active while not allowed**EXPLANATION**

The safeguard stop auto state sent to the safety system is active while the robot is not in safeguard stop auto

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A123 The safeguard stop auto state is inactive while not allowed**EXPLANATION**

The safeguard stop auto state sent to the safety system is inactive while the robot is in safeguard stop auto

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A124 The 3PE stop state is inactive while not allowed**EXPLANATION**

The 3PE stop state sent to the safety system is inactive while the robot is in 3PE stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A125 The 3PE stop state is active while not allowed**EXPLANATION**

The 3PE stop state sent to the safety system is active while the robot is not in 3PE stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A126 The category 2 stop state is inactive while not allowed**EXPLANATION**

The category 2 stop state sent to the safety system is inactive while the robot is in a category 2 stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

**C283A127 The category 2 stop state is active while not allowed****EXPLANATION**

The category 2 stop state sent to the safety system is active while the robot is not in a category 2 stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A128 The category 0 stop state is inactive while not allowed**EXPLANATION**

The category 0 stop state sent to the safety system is inactive while the robot is in a category 0 stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A129 The category 0 stop state is active while not allowed**EXPLANATION**

The category 0 stop state sent to the safety system is active while the robot is not in a category 0 stop

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A130 The safety mode limit is incorrect**EXPLANATION**

The safety mode limit sent to the safety system does not match the safety mode limit in the safety system

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A131 The operational mode is incorrect**EXPLANATION**

The operational mode sent to the safety system does not match the operational mode in the safety system

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A134 Teach pendants without 3PE not supported**EXPLANATION**

The robot only supports teach pendants with a build in 3-Position Enabling Device, but the configured teach pendant doesn't have one

C283A135 Freedrive safety-input disagreement within the safety system**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A136 Brake monitoring not initialized correctly.**EXPLANATION**

Internal setup of brake monitor not setup correctly.

SUGGESTION

Update to the latest software version.

C283A137 Invalid number of dynamic safety clients.

EXPLANATION

The amount of dynamic safety clients requested is higher than the amount of dynamic safety clients allowed.

SUGGESTION

Try the following actions to resolve the issue: (A) Remove unnecessary dynamic safety clients, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C283A143 Tool safety input is configured but the safety configuration version does not support it**EXPLANATION**

The safety configuration version does not support tool safety inputs

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C283A144 Tool safety inputs are configured but the robot type does not support it**EXPLANATION**

The robot type does not support tool safety inputs

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A145 Tool safety inputs are configured but the joint hardware does not support it**EXPLANATION**

The joint hardware does not support tool safety inputs

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A146 Tool safety inputs are configured but the safety input type is not supported**EXPLANATION**

The safety input type is not supported when using the tool safety input

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A147 Tried to get the state of A IO {unsigned} which is not a valid IO**EXPLANATION**

The requested IO is not valid

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C283A148 Tried to get the state of B IO {unsigned} which is not a valid IO**EXPLANATION**

The requested IO is not valid

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C283A149 Tool safety inputs are configured, but the robot configuration does not contain a wrist3 joint**EXPLANATION**

The robot configuration does not contain a wrist3 joint, but tool safety inputs are configured

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Make sure the robot configuration is correct, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A150 Tool safety inputs are configured, but was outside the limits for too long**EXPLANATION**

When the tool safety input are outside the limits, it is neither ON or OFF making it unusable.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Make sure the robot configuration is correct, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A152 Safety plane is configured with IO, but no IO is configured**EXPLANATION**

When safety plane is configured with IO, at least one input or output must be configured

SUGGESTION

Try the following actions to see which resolves the issue: (A) Configure the input or output, (B) Conduct a complete rebooting sequence, (C) Update Software, (D) Contact your local Universal Robots service provider for assistance

C283A153 Safety Plane with IO safety-input disagreement within the safety system**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A154 Safety Plane with IO safety-output disagreement within the safety system**EXPLANATION**

The output signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both outputs are properly connected, (B) Contact your local Universal Robots service provider for assistance

C283A155 The Safety Plane Output is not active**EXPLANATION**

The output signal of a safety plane is not active, when it is expected to be active

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A157 The Safety Plane Output is not inactive**EXPLANATION**

The output signal of a safety plane is not inactive, when it is expected to be inactive

SUGGESTION

Try the following actions to resolve the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A164 Three position enabling stop output is in an incorrect state**EXPLANATION**

Failure to activate the three position enabling stop output. The output is active when signal is low

SUGGESTION

Make sure output is not short circuited to a power supply

C283A165 Three position enabling stop output is in an incorrect state

EXPLANATION

Failure to deactivate the Three position enabling stop output. The output is inactive when signal is high

SUGGESTION

Make sure output is not short circuited to ground

C283A166 No three position enabling stop output is in an incorrect state**EXPLANATION**

Failure to activate the not three position enabling stop output. The output is active when signal is low

SUGGESTION

Make sure output is not short circuited to a power supply

C283A167 No three position enabling stop output is in an incorrect state**EXPLANATION**

Failure to deactivate the not three position enabling stop output. The output is inactive when signal is high

SUGGESTION

Make sure output is not short circuited to ground

C283A168 Lost {unsigned} Tool safety packages in a row**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Check that the Injection-Molding-Machine-Interface is properly connected, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A169 Three position enabling stop output disagreement within the safety system**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A170 Not three position enabling stop output disagreement within the safety system**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A171 Tool safety inputs are configured but not supported**EXPLANATION**

The hardware does not support tool safety inputs

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A172 Received an unexpected tool safety message**EXPLANATION**

The safety system received an unexpected tool safety message

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A173 No tool is configured but tool safety inputs are configured**EXPLANATION**

The safety system is configured to use tool safety inputs, but no tool is configured

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Configure a tool, (C) Update software, (D) Contact your local Universal Robots service provider for assistance



C283A175 Tool type disagreement within the safety system

EXPLANATION

The tool types received by the safety system do not match

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Configure a tool, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C283A177 Tool safety inputs are not supported in this software version

EXPLANATION

The current software version does not support tool safety inputs

SUGGESTION

Try the following actions to see which resolves the issue: (A) Disable tool safety input, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A178 Parking did not begin within the expected time after emergency stop

EXPLANATION

Parking did not begin within the expected time after emergency stop

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A179 The PROFIsafe configuration is obsolete.

EXPLANATION

The safety configuration being applied assumes that the PROFIsafe signals from the robot are active high. This version of the software will use active low for PROFIsafe signals.

SUGGESTION

Try the following actions to resolve the issue: (A) Re-apply the Safety Configuration, (B) Create a new installation, (C) Update the software, (D) Contact your local Universal Robots service provider for assistance

C283A180 The maximum rotation distance in the movement monitor is too large. Max distance: {float} rad.

EXPLANATION

The calculated maximum rotation distance in the movement monitor exceeds acceptable limits.

SUGGESTION

Try the following actions to resolve the issue: (A) Perform a full reboot sequence, (B) Update the software, (C) Contact your local Universal Robots service provider for assistance

C283A181 Mismatch between tool A type and input state

EXPLANATION

The tool type and input state received from Tool A do not match

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A182 Mismatch between tool B type and input state

EXPLANATION

The tool type and input state received from Tool B do not match

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A185 The minimum payload limit must be zero or greater. Actual value is {float}kg

EXPLANATION

A payload with a negative mass is not supported.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A186 The maximum payload must be greater than or equal to the minimum payload**EXPLANATION**

The payload range is invalid and cannot be applied

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A189 Center of gravity sphere radius must be zero or greater. Actual value is {float}m**EXPLANATION**

Radius represents a distance, distances cannot be negative.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A190 A payload mass below the minimum payload limit was received. Actual value is {float}kg**EXPLANATION**

The payload mass is below the minimum limit

SUGGESTION

Make sure there are no script commands that set an invalid payload mass

C283A191 A payload mass above the maximum payload limit was received. Actual value is {float}kg**EXPLANATION**

The payload mass is above the maximum limit

SUGGESTION

Make sure there are no script commands that set an invalid payload mass

C283A192 A Payload center of gravity outside the allowed limits was received**EXPLANATION**

The payload center of gravity is outside the configured limits

SUGGESTION

Make sure there are no script commands that set an invalid payload center of gravity

C283A200 Manual mode high speed output disagreement within the safety system**EXPLANATION**

The output signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A201 The manual mode reduced speed configuration must be greater than zero. Actual value is {float}mm/s**EXPLANATION**

Negative speed is not supported.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance



C283A202 The manual mode reduced speed configuration is above the limit. Actual value is {float}mm/s

EXPLANATION

The manual mode reduced speed is too great.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A204 Robot moved with a speed of {float}mm/s at the tool. This exceeds the manual mode reduced speed limit in the safety settings

EXPLANATION

The tool speed is above the manual mode reduced speed limit.

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements, (C) Check safety settings

C283A205 Robot moved with a speed of {float}mm/s at the elbow. This exceeds the manual mode reduced speed limit in the safety settings

EXPLANATION

The elbow speed is above the manual mode reduced speed limit.

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements, (C) Check safety settings

C283A206 Manual mode high speed output is in an incorrect state

EXPLANATION

Failure to activate the manual mode high speed output. The output is active when signal is low

SUGGESTION

Make sure output is not short circuited to a power supply

C283A207 Manual mode high speed output is in an incorrect state

EXPLANATION

Failure to deactivate the manual mode high speed output. The output is inactive when signal is high

SUGGESTION

Make sure output is not short circuited to ground

C283A208 Parking did not begin within the expected time after safety stop

EXPLANATION

Parking did not begin within the expected time after safety stop

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR service provider for assistance

C283A210 Wrist position limit exceeded

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A211 Scanner Front Input disagreement within the safety system

EXPLANATION

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Update scanner configuration, (C) Contact your local MiR service provider for assistance

C283A212 Scanner Rear Input disagreement within the safety system**EXPLANATION**

The input signals are not switching simultaneously, or are incorrectly connected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure both inputs are properly connected, (B) Update scanner configuration, (C) Contact your local MiR service provider for assistance

C283A213 The freedrive speed configuration must be greater than zero. Actual value is {float}mm/s**EXPLANATION**

Negative speed is not supported.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C283A214 Robot moved with a speed of {float}mm/s at the tool. This exceeds the freedrive speed limit in the safety settings**EXPLANATION**

The tool speed is above the freedrive speed limit.

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements, (C) Check safety settings

C283A215 Robot moved with a speed of {float}mm/s at the elbow. This exceeds the freedrive speed limit in the safety settings**EXPLANATION**

The elbow speed is above the freedrive speed limit.

SUGGESTION

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements, (C) Check safety settings

C283A216 The Wrist moved too fast towards a safety plane**SUGGESTION**

Try each of the following actions (in order) to see which resolves the issue: (A) Ensure the mounted payload and center of gravity matches your installation settings, (B) Slow down movements. You can exceed speed limits if you move the robot too fast in Freedrive, in that case, move the robot slower, (C) Check safety settings

C283A220 Invalid monitoring case {unsigned} was requested for the front scanner**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Update scanner configuration, (B) Contact your local MiR service provider for assistance

C283A221 Invalid monitoring case {unsigned} was requested for the rear scanner**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Update scanner configuration, (B) Contact your local MiR service provider for assistance

C283A222 The linear velocity of {float} m/s is outside the limits of the active scanner monitoring case

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Update scanner configuration, (B) Contact your local MiR service provider for assistance

C283A223 The angular velocity of {float} rad/s is outside the limits of the active scanner monitoring case**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Update scanner configuration, (B) Contact your local MiR service provider for assistance

C283A230 Mismatch on Safety Field Configuration CRC between the safety system and the configuration**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR provider for assistance

C283A231 Audio/visual motion warning violation**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR provider for assistance

2.24. C284 Brake Release

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C284A0 Critical error**EXPLANATION**

A critical error occurred during Brake Release

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Ensure the mounted payload, TCP, and CoG matches your configuration, (C) Contact your local Universal Robots service provider for assistance

C284A21 Brake release count reached limit**EXPLANATION**

The max number of allowed brake releases is reached. The joint can no longer release the brakes.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C284A22 Brake release count is close to the limit, remaining brake releases: {unsigned}**EXPLANATION**

The joint is close to reaching the brake release limit, once reached the joint can no longer release the brakes.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

2.25. C285 Joint Keep-Alive System

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C285A0 Critical error

EXPLANATION

A critical error occurred in the Joint Keep-Alive System

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C285A10 Lost {unsigned} Keep-Alive System message(s) in a row from Safety Control Board-uPA

EXPLANATION

An invalid amount of Keep-Alive System messages have been lost from the Safety Control Board Processor A

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C285A11 Lost {unsigned} Keep-Alive System message(s) in a row from Safety Control Board-uPB

EXPLANATION

An invalid amount of Keep-Alive messages have been lost from the Safety Control Board Processor B

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.26. C286 Motor Controller

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C286A2 PWM margin too small, ticks left: {signed}

EXPLANATION

Motor control failed to meet timing requirements

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C286A4 PWM is not zero when in power off

EXPLANATION

The motor PWM output is not set to zero when in power off.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance



2.27. C287 Saved files

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C287A0 Critical error

EXPLANATION

A critical error occurred during file loading/saving

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C287A1 The file requested (id {unsigned}) is not saved or its loading failed

EXPLANATION

A problem was encountered while trying to load a file in the file system of the joint

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.28. C289 Tool Connector

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C289A1 Too high current detected on Digital Output: {unsigned} high side

EXPLANATION

Tool Digital Output pin has been turned off as a too high source current was detected.

SUGGESTION

Check connections to make sure Tool Digital Output currents are within specified limits.

C289A2 Too high sink current detected on Digital Output: {unsigned} low side

EXPLANATION

Tool Digital Output pin has been turned off as a too high sink current was detected.

SUGGESTION

Check connections to make sure Tool Digital Output sink currents are within specified limits.

C289A4 10 second Average tool IO Current of {float} A is outside of the allowed range.

EXPLANATION

The average current sum of the Tool Connector Power and Digital Output pins is outside of the allowed range.

SUGGESTION

Check connections to make sure tool Digital Output currents are within specified limits.

C289A5 Unable to remove tool Digital Output fault.

EXPLANATION

Unable to remove the overload on tool Digital Output, therefore the robot powered down.

SUGGESTION

Check connections to make sure the Tool Digital Output currents are within specified limits.

C289A6 Current of {float} A on the tool connector supply pins is outside of the allowed range.

EXPLANATION

Too high current on tool connector supply pins

SUGGESTION

Check connections to make sure the Tool Digital Output currents are within specified limits.

C289A7 Current of {float} A on the Digital Output pins is outside of the allowed range.

EXPLANATION

Too high current on tool connector Digital Output pins.

SUGGESTION

Check connections to make sure the Tool Digital Output currents are within specified limits.

C289A8 Current of {float} A on the ground pin is outside of the allowed range.

EXPLANATION

Too high current on tool ground pin

SUGGESTION

Check connections to make sure the Tool Digital Output currents are within specified limits.

C289A9 Current of {float} A on the POWER pin is outside of the allowed range.

EXPLANATION

Too high current on tool power pin

SUGGESTION

Check connections to make sure the Tool Digital Output currents are within specified limits.

C289A10 The M8 digital output sinking current of {float} A exceeded the allowed single pin limit.

EXPLANATION

The M8 digital output sinking current exceeded the allowed single pin limit.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure you are not sinking too much current into the digital output pin, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C289A11 The M8 digital output sinking current of {float} A exceeded the allowed dual pin limit.

EXPLANATION

The M8 digital output sinking current exceeded the allowed dual pin limit.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure you are not sinking too much current into the digital output pins, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C289A12 The M8 digital output 0 sourcing current of {float} A exceeded the allowed limit.

EXPLANATION

The M8 digital output 0 sourcing current exceeded the allowed limit.

C289A13 The M8 digital output 1 sourcing current of {float} A exceeded the allowed limit.

EXPLANATION

The M8 digital output 1 sourcing current exceeded the allowed limit.

C289A14 The zero current of the M8-power current sensor exceeds maximum allowed offset. The measured offset was {float}A

EXPLANATION

The zero current of the M8-power current sensor exceeds maximum allowed offset.

C289A15 The momentary M8-power current of {float}A exceeded the allowed limit.

**EXPLANATION**

The momentary current drawn from the M8-power connector exceeded the allowed limit.

C289A16 The M8-power current exceeded the the allowed limit of {float}A for more than 2 seconds during the last 10 seconds**EXPLANATION**

The M8-power current exceeded the the allowed limit for more than 2 seconds during the last 10 seconds

C289A17 The zero current of the IO M8 power current sensor exceeds maximum allowed offset. The measured offset was {float}A**EXPLANATION**

The zero current of the IO M8 power current sensor exceeds maximum allowed offset.

C289A18 The zero current of the IO M8 digital output current sensor exceeds maximum allowed offset. The measured offset was {float}A**EXPLANATION**

The zero current of the IO M8 digital output current sensor exceeds maximum allowed offset.

2.29. C292 Online RAM test

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C292A0 Critical error**EXPLANATION**

A critical error occurred during RAM test

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.30. C294 ADC

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C294A0 Critical error**EXPLANATION**

A critical error occurred in the ADC driver

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C294A12 The single ended ADC calibration timed out**EXPLANATION**

The single ended ADC calibration timed out, thus the found calibration factor is not valid

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C294A16 The joint failed during recalibration of the ADC offset**EXPLANATION**

The joint failed during recalibration of the ADC offset, because there was too much noise on the measurements

SUGGESTION

Make sure that the robot stands completely still and there is no external electrical noise sources. If the problem persists contact technical support

C294A19 The motor moves too much during ADC offset sampling, the motor position has a standard deviation of {float} [rad] during the sampling**EXPLANATION**

The motor moves too much during ADC offset sampling, and the ADC measurements are therefore not sane

SUGGESTION

Make sure the robot is at complete standstill and it is not in a vibrating environment during startup

2.31. C295 PCB

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C295A0 Wrong PCB type ({hex})**EXPLANATION**

The printed circuit board is defective

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.32. C296 Start up check

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C296A0 Critical error**EXPLANATION**

A critical error occurred during startup

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C296A1 SCB IO failed to power on**SUGGESTION**

Ensure the IO Power Connector on the Safety Control Board is connected to the 24VDC

C296A2 One or more Motor phases is short circuited to ground. Diagnostic data: {hex}



EXPLANATION

Bit 0 high indicates short circuit

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A3 Motor Indication Signal does not work. Diagnostic data: {hex}

EXPLANATION

Bit 1 low indicates that the diagnostics feature is broken

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A4 Phase 1 is not connected. Diagnostic data: {hex}

EXPLANATION

Bit pattern 0b0110 indicates that phase 1 is not connected

SUGGESTION

Make sure all Motor phases are connected correctly. Contact your local Universal Robots service provider for assistance.

C296A5 Phase 2 is not connected. Diagnostic data: {hex}

EXPLANATION

Bit pattern 0b1010 indicates that phase 2 is not connected

SUGGESTION

Make sure all Motor phases are connected correctly. Contact your local Universal Robots service provider for assistance.

C296A6 Phase 3 is not connected. Diagnostic data: {hex}

EXPLANATION

Bit pattern 0b0010 indicates that phase 3 is not connected

SUGGESTION

Make sure all Motor phases are connected correctly. Contact your local Universal Robots service provider for assistance.

C296A7 Motor test results were invalid. Diagnostic data: {hex}

EXPLANATION

Bit0 high indicates short circuit. Bit1 is phase 3 test. Bit2 is phase 2 test. Bit3 is phase 1 test

SUGGESTION

Make sure all Motor phases are connected correctly. Contact your local Universal Robots service provider for assistance.

C296A9 Robot Voltage was present during self-diagnostics

EXPLANATION

Robot Voltage rose above acceptable levels before both processors powered it on

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A10 Time out during self-diagnostics

EXPLANATION

A processor timed out while waiting for the other processor to finish self-diagnostics

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A11 Data was received while trying to disable communication

EXPLANATION

A validation that the communication could be suppressed failed to prevent a message from passing through.

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A12 Sequence number did not match expected sequence

EXPLANATION

After communication was disabled and reenabled the sequence number did not match what was expected

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A15 Interval between messages did not match expectations

EXPLANATION

After communication was disabled and reenabled the message interval did not match what expectations

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A23 Cross-monitoring data was invalid for too long while booting

EXPLANATION

uB timed out while waiting for cross-monitoring data to agree before changing from Booting to Idle state

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A25 Diagnostics module was initialized with a NULL pointer, for a required callback function

EXPLANATION

Diagnostics module was initialized with a null pointer, for rotate com buffer callback

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A33 Timeout while waiting for system to be ready to go to idle

EXPLANATION

Requirements for going to idle state were not met within the expected time frame

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A36 The device initialization has failed.

EXPLANATION

The device initialization has failed. this can be caused by hardware failure

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Contact your local Universal Robots service provider for assistance.

C296A37 Robot Safe Torque Off Failed during self-diagnostics

EXPLANATION

Robot failed while performing Safe Torque Off self-diagnostics, which can be caused by a failure in the Safe Torque Off circuitry.

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C296A44 24V voltage is not operating at an acceptable level, the actual value is {float}V

EXPLANATION

The 24V voltage is not within the acceptable operational range.

SUGGESTION

Check the 24V power supply and connections, and contact your local Universal Robots service provider for assistance if the issue persists.



2.33. C297 Joint validation

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C297A0 Critical error

EXPLANATION

A critical error occurred during Joint validation

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check that SD card and robot type match, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C297A11 The Robot arm does not match the Control Box

EXPLANATION

One or more joints do not match stored Robot Configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check that SD card and robot type match, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C297A13 Joint type received from joint {unsigned} is invalid

EXPLANATION

A joint in the arm has reported unsupported joint type.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C297A14 Joint type received from joint {unsigned} is invalid.

EXPLANATION

The safety system has detected a mismatch between the detected and the configured joint type.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance, (D) Change the joint to the correct joint type.

C297A15 The type detected by the joint was {unsigned}.

EXPLANATION

Joint type received from the joint.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance.

C297A16 The expected joint type specified in the robot configuration was {unsigned}.

EXPLANATION

Joint type configured, and thereby expected from the joint.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance.

C297A17 The safety processors on joint {unsigned} disagree on the joint type.

EXPLANATION

There is a mismatch between the joint size type detected by processor A and processor B.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance.

C297A18 Joint type detected by processor A: {unsigned}.**EXPLANATION**

Processor A has detected the joint type during a type mismatch

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance.

C297A19 Joint type detected by processor B: {unsigned}.**EXPLANATION**

Processor B has detected the joint type during a type mismatch.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance.

2.34. C298 Hand protection

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C298A0 Tool is too close to the lower arm: {float_2_4} meter.**EXPLANATION**

The tool is too close to the lower Robot arm

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check wrist position, (B) Verify mounting, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

2.35. C299 Tool communication

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C299A0 Communication error detected**EXPLANATION**

A problem with the Tool communication was detected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C299A3 RX framing error

**EXPLANATION**

Framing error detected on received data

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check external equipment connection, (B) Verify the communication configuration matches the hardware, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C299A4 RX Parity error**EXPLANATION**

Parity error detected on received data

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check external equipment connection, (B) Verify the communication configuration matches the hardware, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

2.36. C301 Safety message monitor

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C301A0 Critical error**EXPLANATION**

A critical error occurred in safety message monitoring

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.37. C302 Tool Configuration

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C302A1 Invalid Robot Type {unsigned}**EXPLANATION**

The tool received an invalid robot type

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

2.38. C303 System status

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C303A0 Critical error

EXPLANATION

A critical system error occurred

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C303A6 Reset caused by independent watchdog

EXPLANATION

The FW has hit a deadlock and therefore reset by watchdog

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.39. C304 Self monitoring

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C304A0 Critical error

EXPLANATION

A critical error occurred in physical, logical, and temporal monitoring (PLATM)

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C304A3 Close to the gearbox shear limit. Encoders disagree {float} [rad] on the Joint position

EXPLANATION

The Joint acceleration or deceleration is too high, or there is a mechanical problem in the gear related to encoder mounting.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce acceleration in the user program, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Replace Joint if necessary, (E) Contact your local Universal Robots service provider for assistance

C304A4 Either the encoder was inappropriately mounted, or the gearbox is loose or broken. Difference between the encoders is {float} [rad].

EXPLANATION

Mechanical problem in gear related to encoder mounting.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce acceleration in the user program, (B) Check TCP, Payload, and Cog, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Replace Joint if necessary, (F) Contact your local Universal Robots service provider for assistance

C304A6 Motor phase {unsigned}'s resistance is too high.

EXPLANATION

The lead/connector is broken, or the Motor phase lead has become disconnected or loose.

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.40. C305 Robot Power Control

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C305A0 Critical error

EXPLANATION

A critical error in power control module

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C305A1 Power supply voltage too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check for loose connections, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C305A2 Robot cable not connected

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the cable between Control Box and Robot Arm is correctly connected and it has no damage, (B) Check for loose connections, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C305A3 Short circuit in Robot detected or the wrong Robot is connected to the Control Box.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check for loose connections, (B) Make sure the cable between Control Box and Robot Arm is correctly connected and it has no damage, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C305A4 Robot voltage rising slower than expected

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the power supply, (B) Verify Control Box and Robot Arm are correctly paired, (C) Contact your local Universal Robots service provider for assistance

C305A6 Power supply voltage too high: {float} V

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check for loose connections, (B) Make sure the cable between Control Box and Robot Arm is correctly connected and it has no damage, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C305A8 The Robot Voltage is too high ({float})V when powering on the Robot

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C305A9 The Power State was not OFF ({unsigned}) when trying to power on the Robot

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C305A11 The power to the robot arm was not removed fast enough after violation

C305A12 Unexpected energy eater type

EXPLANATION

The detected energy eater type was different from the expected type. This may be due to a wrong energy eater type being installed in the control box, or that the connections to the energy eater are damaged.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check the connection between the control board and the energy eater, (C) Check if the correct energy eater is installed (D) Update software, (E) Contact your local Universal Robots service provider for assistance

2.41. C306 Joint

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C306A0 Critical error

EXPLANATION

A critical error occurred in a Joint

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C306A1 Not stopping fast enough

EXPLANATION

Joint was unable to come to a full stop fast enough.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Either an incorrect payload is mounted, or an external force is pushing the robot, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C306A2 Velocity failed to pass sanity check

EXPLANATION

The received Joint velocity target is invalid.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Adjust the robot program to reduce peak acceleration and torques, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C306A3 Acceleration failed to pass sanity check

EXPLANATION

The received Joint acceleration target is invalid.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Adjust the robot program to reduce peak acceleration and torques, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C306A9 Joint moved more than allowable limit



EXPLANATION

Potential mechanical failure of the joint's brakes

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C306A15 Joint moved too far while it should be stationary

EXPLANATION

The robot moved more than allowed during parking.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C306A16 A timeout occurred while in violation parking

EXPLANATION

The joint reached the timeout limit while in violation parking.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C306A17 The joint parking procedure violated max allowed execution time of {float} [ms].

EXPLANATION

The joint reached the timeout limit while in parking.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C306A19 Illegal state transition, the joint entered parking while motorcontrol was disabled.

EXPLANATION

The parking procedure cannot execute without motorcontrol enabled.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C306A20 Trying to power off from violation while motor control is enabled.

EXPLANATION

Tried to skip parking procedure while motor control was enabled.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C306A21 Unexpected joint type ({hex})

EXPLANATION

The joint has identified its type incorrectly

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C306A22 Trying to enter idle brake failure while motor control is enabled.

EXPLANATION

The joint should not enter idle brake failure with motor control enabled.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C306A24 Motor unsupported by the Critical Fault brake handler**EXPLANATION**

The joint cannot be allowed to brake release, as it has no defined Critical Fault braking profile for its motor type

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

2.42. C309 Keep-Alive

You may find the following links helpful in addressing this issue:

- [How to perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C309A0 Critical error**EXPLANATION**

A critical error occurred in the Keep-Alive

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C309A1 Message with wrong sequence received from SCB-uPA.**EXPLANATION**

The Keep-Alive message received from the Safety Control Board Processor A has an invalid sequence number

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C309A4 Message with wrong sequence received from SCB-uPB.**EXPLANATION**

The Keep-Alive message received from the Safety Control Board Processor B has an invalid sequence number

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C309A10 Lost {unsigned} Keep-Alive message(s) in a row from Safety Control Board-uPA**EXPLANATION**

An invalid amount of Keep-Alive messages have been lost from the Safety Control Board Processor A

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C309A11 Lost {unsigned} Keep-Alive message(s) in a row from Safety Control Board-uPB**EXPLANATION**

An invalid amount of Keep-Alive messages have been lost from the Safety Control Board Processor B

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C309A14 Invalid command received in Keep-Alive message from SCB-uPA.**EXPLANATION**

The Keep-Alive message received from the Safety Control Board Processor A has an invalid alive command

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C309A15 Invalid command received in Keep-Alive message from SCB-uPB.**EXPLANATION**

The Keep-Alive message received from the Safety Control Board Processor B has an invalid alive command

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.43. C310 Joint Temperature Manager

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C310A0 Critical error**EXPLANATION**

A critical error occurred in the joint temperature manager

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.44. C312 Data validation

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C312A0 Critical error**EXPLANATION**

A critical error occurred during data validation

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for ESD noise, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

2.45. C314 SPI IO

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C314A0 Critical error

EXPLANATION

A critical error occurred related to IO

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for IO connections, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C314A5 Expected OSSD pulse were not detected on CO{unsigned}

EXPLANATION

The generated OSSD pulses on the safety output were not seen during readback

SUGGESTION

Try any of the following actions to resolve the issue: (A) Verify safety output is not connected to any power supply or another safety output, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C314A6 An unexpected OSSD pulse was detected on CO{unsigned}

EXPLANATION

An OSSD pulse was detected on the safety output readback, but was not generated by the hardware

SUGGESTION

Try any of the following actions to resolve the issue: (A) Verify safety output is not connected to ground or another safety output, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C314A7 The IO voltage is missing or below threshold

EXPLANATION

The IO voltage is missing or below the threshold, a fuse may be blown or removed

SUGGESTION

Ensure the IO Power Connector on the Safety Control Board is connected to the 24VDC and the fuse is in working condition

C314A8 The configurable safety outputs were not low when expected

EXPLANATION

The configurable safety outputs were not low when expected

SUGGESTION

Try any of the following actions to resolve the issue: (A) Verify safety output is not connected to any power supply or another safety output, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C314A11 Configurable IO circuit asserted its alert signal

EXPLANATION

Read the alert status register for specific error information

SUGGESTION

Try any of the following actions to resolve the issue: (A) Verify safety output is not connected to any power supply or another safety output, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C314A12 Could not allocate RX and/or TX buffer

EXPLANATION

The SPI DMA buffers are frequently rotated (freed and reallocated). The allocator returned a null pointer.

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C314A13 Configurable IO circuit reports invalid CRC or clock count error**EXPLANATION**

Transmitted frames must be exactly 32 bits in length and contain a valid 8-bit CRC

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for ESD noise or interference on the SPI bus, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C314A18 Timed out while waiting for conversion to complete**EXPLANATION**

A timeout occurred while waiting for the "ADC ready" pin to assert

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C314A19 Timed out while waiting for a transaction to complete**EXPLANATION**

A timeout occurred while waiting for an "SPI transaction complete" callback

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.46. C315 Watchdog

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C315A0 Self-test failed**EXPLANATION**

The system watchdog is not working as expected

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C315A1 Verification found an error at index {unsigned}**EXPLANATION**

The watchdog timing and check-in counts are monitored, if these are outside the limits the system faults.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C315A2 Checked in at {float_1_3} ms which is outside the permitted window.**EXPLANATION**

The watchdog check-in was outside the permitted window.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C315A5 Keeper module ran out of space.**EXPLANATION**

The number of allowed checkpoints has exceeded the allowed limit.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C315A6 Trying to register checkpoint with uninitialized keeper.**EXPLANATION**

Trying to register a checkpoint, before the watchdog keeper is initialized is not allowed.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C315A7 Keeper failed to initialize, as it was already initialized.**EXPLANATION**

The watchdog keeper cannot be initialized multiple times.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C315A28 The watchdog configuration could not be applied**EXPLANATION**

The watchdog could not be configured as requested.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.47. C316 MCU

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C316A0 Unknown ID**EXPLANATION**

The Microcontroller Identifier does not match an expected value

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C316A1 This version of the firmware is obsolete and needs to be updated**EXPLANATION**

The firmware in the robot is too old and needs to be updated

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance



C316A9 The device is not whitelisted

EXPLANATION

The software is not allowed to run on this microcontroller

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.48. C328 Transceiver miscellaneous

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C328A2 Flash device is not supported, JEDEC data for device is: {hex}

EXPLANATION

The flash device's JEDEC ID does not match a supported flash device

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.49. C329 AXI STREAM

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C329A3 Channel {unsigned} not configured

EXPLANATION

Attempting to send data on an unconfigured interface.

SUGGESTION

Contact your local Universal Robots service provider for assistance.

C329A4 Write failed, {unsigned} messages dropped

EXPLANATION

Tried to transmit more data than the Axi-Stream could consume, and messages where lost.

SUGGESTION

Contact your local Universal Robots service provider for assistance.

2.50. C330 IMMI IO

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C330A1 Injection-Molding-Machine-Interface E-Stop output readback does not match produced value: {hex}

EXPLANATION

First byte: produced value, second byte: read value

C330A2 Injection-Molding-Machine-Interface Moulding Area Free output readback does not match produced value: {hex}

EXPLANATION

First byte: produced value, second byte: read value

C330A3 Injection-Molding-Machine-Interface 24V IO voltage outside acceptable range

EXPLANATION

The voltage measured on the 24V IO rail is lower than expected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the fuses on the Injection-Molding-Machine-Interface. (B) Verify there are no short circuits on the 24V IO connectors. (C) Contact your local Universal Robots service provider for assistance.

C330A4 Injection-Molding-Machine-Interface 48V voltages outside acceptable range

EXPLANATION

The voltages measured on the 48V rails are lower than expected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the fuses on the Injection-Molding-Machine-Interface. (B) Verify there are no short circuits on the IO connectors. (C) Contact your local Universal Robots service provider for assistance.

2.51. C331 Friction model

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C331A0 Critical error

EXPLANATION

A critical error occurred in the friction model module

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C331A1 The velocity of the friction model is outside the limits,

EXPLANATION

The velocity used for the friction model is outside the limits.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C331A2 The current of the friction model is outside the limits

EXPLANATION

The current used for the friction model is outside the limits.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C331A3 The temperature of the friction model is outside the limits

EXPLANATION

The temperature used for the friction model is outside the limits.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C331A4 The element count of the friction model is wrong

**EXPLANATION**

The element count used for the friction model is wrong.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C331A5 The calibration file for the friction model validation was not found**EXPLANATION**

The calibration file used to validate the friction model was not found.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C331A6 The limit data for the friction model was not found**EXPLANATION**

The limit data used for the friction model was not found.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.52. C332 Servo configuration

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C332A0 Critical error**EXPLANATION**

A critical error occurred in the servo configuration module

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.53. C333 File message

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C333A0 Critical error**EXPLANATION**

A critical error occurred in the file message module

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.54. C334 Robot deviated from constrained axes while in Constrained Freedrive.

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

2.55. C337 Control parameters

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C337A0 Critical error

EXPLANATION

A critical error occurred during setup of the control parameters

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.56. C338 PROFIsafe

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C338A0 Critical error

EXPLANATION

A critical error occurred in the PROFIsafe logic

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C338A6 A PROFIsafe message was received, but there is no valid configuration

EXPLANATION

A PROFIsafe message was received before a valid safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the PROFIsafe Host settings and review the robot safety configuration, see PROFIsafe (B) Do not start a PROFIsafe host before the robot is ready (C) Conduct a complete rebooting sequence (D) Contact your local Universal Robots service provider for assistance

C338A7 A PROFIsafe message was received and PROFIsafe was disabled.

EXPLANATION

A PROFIsafe message was received while PROFIsafe is disabled in the safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Verify the PROFIsafe Host settings and review the robot safety configuration, see PROFIsafe (B) Do not start a PROFIsafe host before the robot is ready (C) Conduct a complete rebooting sequence (D) Contact your local Universal Robots service provider for assistance



C338A8 The robot rejected the PROFIsafe F-Parameter set

EXPLANATION

A PROFIsafe F-Parameterset was received from the PLC but was not accepted

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the diagnosis messages from the PLC and set a valid F-Parameter configuration, (B) Conduct a complete rebooting sequence (C) Contact your local Universal Robots service provider for assistance

C338A11 An iPar CRC mismatch has been detected

EXPLANATION

The iPar CRC received from the PLC does not match the Safety checksum

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check if the iPar CRC set in the PLC matches the Safety checksum, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C338A12 An error occurred during PROFIsafe communication

EXPLANATION

The CRC over the cyclic telegram does not match the expected value

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the connection and cable to the PLC, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C338A13 The PROFIsafe watchdog timed out

EXPLANATION

The PROFIsafe communication timed out because no message has been received in the expected time

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the connection and cable to the PLC, (B) Check and adjust the watchdog timeout, (C) Conduct a complete rebooting sequence (D) Contact your local Universal Robots service provider for assistance

C338A14 The PROFINet provider status is bad

EXPLANATION

The PROFINet provider reports a failure or is not reachable

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the connection and cable to the PLC, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C338A17 Destination address mismatch

EXPLANATION

The PROFIsafe destination address received from the PROFIsafe host does not match the destination address in the robot safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure that the destination address in the safety configuration matches the destination address configured on the PROFIsafe host, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C338A18 Destination address invalid

EXPLANATION

The PROFIsafe destination address received from the PROFIsafe host is not valid

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure that the destination address configured on the PROFIsafe host is not set to 0 or 0xFFFF, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C338A19 Source address mismatch

EXPLANATION

The PROFIsafe source address received from the PROFIsafe host does not match the source address in the robot safety configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure that the source address in the safety configuration matches the source address configured on the PROFIsafe host, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C338A20 Source address invalid

EXPLANATION

The PROFIsafe source address received from the PROFIsafe host is not valid

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure that the source address configured on the PROFIsafe host is not set to 0 or 0xFFFF, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

C338A21 Wrong PROFIsafe submodule selected

EXPLANATION

The PROFIsafe submodule id received from the PROFIsafe host does not match the control operational mode setting in the Polyscope PROFIsafe configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the selected submodule in the PROFIsafe host and the control operational mode setting in the Polyscope PROFIsafe configuration, (B) Conduct a complete rebooting sequence, (C) Contact your local Universal Robots service provider for assistance

2.57. C339 Cross communication

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C339A0 Critical error

EXPLANATION

A critical error occurred in the cross communication module

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.58. C340 Energy Monitoring

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C340A0 Idle power consumption too high

EXPLANATION

The system is drawing more power than expected while idle

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure supply is not delivering more than 48V, (B) Check Energy Eaters cable and connections, (C) Check Energy Eater, (D) Update software, (E) Contact your local Universal Robots service provider for assistance



C340A1 Energy surplus shutdown

EXPLANATION

The power supply is sending energy to the energy eater

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure supply is not delivering more than 48V, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C340A2 Energy burst period exceeded, average power: {float}W

EXPLANATION

Energy has been sent to the energy eater longer than allowed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure supply is not delivering more than 48V, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C340A3 The Energy Removal Device peak power exceeded

EXPLANATION

The absolute limit for the Energy Removal Device has been exceeded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure supply is not delivering more than 48V, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C340A4 The Energy Removal Device drew too much power over a period

EXPLANATION

The power drawn by the Energy Removal Device over a period has exceeded the allowed limit

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure supply is not delivering more than 48V, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.59. C341 Motor encoder

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C341A0 Critical error

EXPLANATION

A critical communication error occurred

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A9 The scheduled motor encoder SPI package transfer did not complete before the deadline.

EXPLANATION

A timing error occurred in the robot firmware.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A10 CRC error in CH0 transfer, frame: {hex}

EXPLANATION

A SPI package from the motor encoder was corrupted during transfer.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the application is not exceeding the capabilities of the robot (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A22 Detailed status error: Signal clipping, strong external magnetic field is present.**EXPLANATION**

The presence of a strong external magnetic field, caused signal clipping on the motor encoders signal.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the application is not exceeding the capabilities of the robot (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A23 Detailed status warning: Signal amplitude too high. The read head is too close to the ring.**EXPLANATION**

The motor encoder read head is too close to the ring or an external magnetic field is present

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the application is not exceeding the capabilities of the robot (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A24 Detailed status error: ASIC was reset, external RFI glitch caused encoder to reset.**EXPLANATION**

External RFI glitch caused encoder to reset.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Try to remove any obvious sources of strong RFI (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A25 Detailed status error: ASIC synchronization lost, external RFI glitch caused malfunction.**EXPLANATION**

External RFI glitch caused encoder to loose synchronization.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Try to remove any obvious sources of strong RFI (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A26 Detailed status error: Encoder not configured properly.**EXPLANATION**

Incorrect motor encoder configuration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A27 Detailed status warning: Signal amplitude low. The distance between the read head and the ring is too large.**EXPLANATION**

The distance between the read head and the ring is too large

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Make sure the application is not exceeding the capabilities of the robot (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A28 Detailed status error: Signal lost. The read head is out of alignment with the ring or the ring is damaged.**EXPLANATION**

The read head is out of alignment with the ring or the ring is damaged

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the application is not exceeding the capabilities of the robot (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A30 Detailed status error: Power supply error. The read head power supply voltage is out of specified range.**EXPLANATION**

The voltage supplied for the motor encoder is outside allowed specifications

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the application is not exceeding the capabilities of the robot (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A31 Detailed status error: System error. Malfunction inside the circuitry or inconsistent calibration data is detected.**EXPLANATION**

A system error occurred inside the motor encoder.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A32 Detailed status error: Magnetic pattern error. A stray magnetic field is present or metal particles are present between the read head and the ring or radial positioning between the read head and the ring is out of tolerances.**EXPLANATION**

Motor encoder was unable to correctly decode the position, possible causes could be (A) Stray magnetic fields (B) Metal particles between read head and ring (C) Installation tolerances is outside specification.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the application is not exceeding the capabilities of the robot (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A33 Detailed status error: Acceleration too high.**EXPLANATION**

Motor encoder reported that the acceleration of the encoder disc was outside allowed specifications.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the application is not exceeding the capabilities of the robot (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C341A65 Latest parsed motor encoder position was too old.**EXPLANATION**

The latest parsed SPI package had a timestamp that was too old, indicating either a timing glitch or corrupted data.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A86 An error occurred while reading the encoder firmware version.**EXPLANATION**

An error occurred while reading the encoder firmware version.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A87 The encoder firmware version is not valid.**EXPLANATION**

The encoder firmware version is not valid.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A88 A reading of the register {hex} was attempted which is not valid for the present encoder**EXPLANATION**

A read from a register that is invalid for the encoder was attempted

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A89 A write of the register {hex} was attempted which is not valid for the present encoder**EXPLANATION**

A write to a register that is invalid for the encoder was attempted

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A90 Tried to access register {hex}, which is not valid for the present encoder**EXPLANATION**

A register that is invalid for the encoder was accessed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A92 The encoder reported that it has not performed a self-calibration**EXPLANATION**

The encoder reported that it has not performed a self-calibration

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C341A95 Detailed status error: System error. Malfunction inside the circuitry.**EXPLANATION**

A system error occurred inside the motor encoder. To reset the System error bit try to cycle the power supply.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the motor-encoder PCBA - Contact your local Universal Robots service provider for assistance



C341A96 Detailed status error: System error. Inconsistent calibration data detected.

EXPLANATION

A system error occurred inside the motor encoder. To reset the System error bit try to cycle the power supply.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the motor-encoder PCBA - Contact your local Universal Robots service provider for assistance

2.60. C342 Motor Parameters

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C342A0 Module initialization error

EXPLANATION

An error occurred during initialization of the Motor Parameters Module, a critical dependency could not be satisfied.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C342A1 Motor datasheet unavailable

EXPLANATION

A missing or invalid motor datasheet, has caused the initialization of the motor parameters module to fail.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C342A2 Loading of the motor parameters file failed.

EXPLANATION

A missing or invalid motor parameters file, has caused the initialization of the motor parameters module to fail.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.61. C343 Joint Configuration

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C343A0 Critical error

EXPLANATION

A critical error occurred in the joint configuration module

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A1 Failed to access the joint general datasheet, as the datasheet has not been loaded

EXPLANATION

The joint type may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A2 Failed to access the joint gear datasheet, as the datasheet has not been loaded

EXPLANATION

The gear type in this joint may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A3 Failed to access the joint motor datasheet, as the datasheet has not been loaded

EXPLANATION

The motor type in this joint may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A4 Failed to access the joint PCBA datasheet, as the datasheet has not been loaded

EXPLANATION

The PCBA in this joint may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A5 Failed to load the joint general datasheet

EXPLANATION

The joint type may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A6 Failed to load the joint gear datasheet

EXPLANATION

The gear type in this joint may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A7 Failed to load the joint motor datasheet

EXPLANATION

The motor type in this joint may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A8 Failed to load the joint PCBA datasheet

**EXPLANATION**

The PCBA in this joint may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A24 Unsupported joint type**EXPLANATION**

The joint type in this joint is not supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A25 Unsupported motor type**EXPLANATION**

The motor type in this joint is not supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A33 The joint have the gear type {unsigned} installed, which is not allowed**EXPLANATION**

The gear installed in the joint is not an allowed type

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C343A34 The joint have the motor type {unsigned} installed, which is not allowed**EXPLANATION**

The motor installed in the joint is not an allowed type

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C343A41 The motor type could not be read from the hardware configuration**EXPLANATION**

Required motor hardware information could not be retrieved from the hardware configuration manager.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A42 Failed to access the joint type, as the joint type is still undetermined**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A43 The gear type could not be read from the hardware configuration**EXPLANATION**

Required gear hardware information could not be retrieved from the hardware configuration manager.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A46 Failed to determine original joint type due to invalid PCB type {unsigned}

EXPLANATION

The joint failed to determine the joint type it originally identified during calibration (selftest) due to an invalid PCB type. This may result in incorrect calibration being applied.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A47 Failed to determine original joint type due to missing motor type information**EXPLANATION**

The joint failed to determine the joint type it originally identified during calibration (selftest) due to missing motor type information. This may result in incorrect calibration being applied.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A48 Failed to determine joint type**EXPLANATION**

The hardware in this joint may not be supported by the current firmware, possibly due to a firmware downgrade

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C343A50 The loaded firmware is too old to work with the calibration data stored in the joint**SUGGESTION**

Update to a newer software release

C343A51 The loaded firmware is too old to fully utilize the calibration data stored in the joint**EXPLANATION**

The joint works normally, but its calibration has improvements this firmware does not utilize.

SUGGESTION

Update to a newer software release to get full joint performance

2.62. C347 Network Map

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C347A4 More robot cable extenders detected than are supported.**EXPLANATION**

It is not supported to have multiple robot cable extenders

SUGGESTION

Try the following actions to see which resolves the issue: (A) Remove the extra robot cable extenders so only one is connected, (B) Update Software, (C) Contact your local Universal Robots service provider for assistance

C347A5 The far end of the robot cable extender was not detected.**EXPLANATION**

Only the near end of the robot cable extender was detected, while the far end is missing.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure that the robot cable extender is properly connected and that the cable is not damaged, (B) Update Software, (C) Contact your local Universal Robots service provider for assistance



2.63. C348 External Axes Bus

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C348A1 Receive error

EXPLANATION

Failed to receive data on External Axes Bus

SUGGESTION

Axis Group feature will be disabled until you conduct a complete rebooting sequence. If problem persists, contact your local Universal Robots service provider for assistance.

C348A2 Send error

EXPLANATION

Failed to send data on External Axes Bus

SUGGESTION

Axis Group feature will be disabled until you conduct a complete rebooting sequence. If problem persists, contact your local Universal Robots service provider for assistance.

C348A3 Target joints calculation failed

EXPLANATION

Failed to calculate target joints for External Axes Bus

SUGGESTION

Try the following actions to see which resolves the issue: (A) Restart UR program, (B) Check each external axis in program for invalid parameters, (C) Conduct a complete rebooting sequence, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C348A4 An enabled external axis at index {unsigned} is no longer receiving drive status updates

EXPLANATION

The EtherCAT Master is no longer sending the Controller status updates for the servo drive of an enabled external axis

SUGGESTION

Try the following actions to see which resolves the issue: (A) Ensure the MotionPlus Service is running, (B) Ensure the EtherCAT Master has been configured for the axis's drive, (C) Ensure the EtherCAT Master is running, (D) Contact your local Universal Robots service provider for assistance

2.64. C349 Base light

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C349A0 Length check error, incorrect length detected for base light ring: {unsigned}

EXPLANATION

The detected length of the base light ring is not what is expected.

SUGGESTION

There is an issue validating the base light. If problem persists, contact your local Universal Robots service provider for assistance.

C349A2 Current for the red color was {float} A, which is outside the allowed range

EXPLANATION

The detected current of the red color in the base light is outside the allowed range.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Perform a complete rebooting sequence, (B) Update software, (C) Check if the connection between the light ring and the PCB is correct, (D) Replace The light ring, (E) Contact your local Universal Robots service provider for assistance

C349A3 Current for the green color was {float} A, which is outside the allowed range**EXPLANATION**

The detected current of the green color in the base light is outside the allowed range.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Perform a complete rebooting sequence, (B) Update software, (C) Check if the connection between the light ring and the PCB is correct, (D) Replace The light ring, (E) Contact your local Universal Robots service provider for assistance

C349A4 Current for the blue color was {float} A, which is outside the allowed range**EXPLANATION**

The detected current of the blue color in the base light is outside the allowed range.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Perform a complete rebooting sequence, (B) Update software, (C) Check if the connection between the light ring and the PCB is correct, (D) Replace The light ring, (E) Contact your local Universal Robots service provider for assistance

C349A5 Current for the yellow color was {float} A, which is outside the allowed range**EXPLANATION**

The detected current of the yellow color in the base light is outside the allowed range.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Perform a complete rebooting sequence, (B) Update software, (C) Check if the connection between the light ring and the PCB is correct, (D) Replace The light ring, (E) Contact your local Universal Robots service provider for assistance

2.65. C350 Failed brake system

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C350A0 Joint position drifted more than allowed**EXPLANATION**

The joint position has drifted more than allowed while in the failed brake state

SUGGESTION

Contact your local Universal Robots service provider for assistance

C350A1 State transition is not permitted with a failed brake system**EXPLANATION**

The joint brake system has failed so the joint is not permitted to enter a state where it can be moved

SUGGESTION

Contact your local Universal Robots service provider for assistance

C350A3 Failure detected in brake system**EXPLANATION**

The brake system failed to engage



2.66. C351 Robot configuration

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C351A0 Unexpected tool detected

EXPLANATION

Tool was detected in the arm, but not in the robot configuration.

SUGGESTION

Update robot configuration to match attached hardware.

C351A1 Unexpected {deviceName} joint detected

EXPLANATION

Joint was detected in the arm, but not in the robot configuration.

SUGGESTION

Update robot configuration to match attached hardware.

C351A2 Unexpected base light detected

EXPLANATION

Base light was detected in the arm, but not in the robot configuration.

SUGGESTION

Update robot configuration to match attached hardware.

C351A3 Robot cable extender detected, but not configured.

EXPLANATION

A robot cable extender was detected, but has not been configured.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Configure the robot cable extender, (B) Remove robot cable extender if it was not connected intentionally, (C) Update Software, (D) Contact your local Universal Robots service provider for assistance

C351A4 Robot cable extender detected, but configuration is corrupted.

EXPLANATION

A robot cable extender was detected, but the configuration is corrupted.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure that the robot cable extender is configured properly, (B) Update Software, (C) Contact your local Universal Robots service provider for assistance

C351A5 Robot cable extender configured, but not detected.

EXPLANATION

A robot cable extender was configured, but was not detected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Unconfigure the robot cable extender, (B) Connect the robot cable extender if it was intended to be connected, (C) Update Software, (D) Contact your local Universal Robots service provider for assistance

C351A6 Unsupported cable type for robot cable extender configured.

EXPLANATION

An unsupported cable type for the robot cable extender was configured.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Configure a supported robot cable extender type, (B) Remove robot cable extender if it was not connected intentionally, (C) Update Software, (D) Contact your local Universal Robots service provider for assistance

C351A7 Detected cable type for robot cable extender is not supported.

EXPLANATION

The detected cable type for the robot cable extender is not supported.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Connect a supported cable type, (B) Update Software, (C) Contact your local Universal Robots service provider for assistance

C351A8 Detected cable type for robot cable extender is incompatible with configured type.

EXPLANATION

The detected cable type for the robot cable extender is incompatible with the configured cable type.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Configure the robot cable extender with the correct cable type, (B) Connect the intended cable between the cable extenders (C) Update Software, (D) Contact your local Universal Robots service provider for assistance

C351A9 Detected cable type for robot cable extender does not match configured type.

EXPLANATION

The detected cable type for the robot cable extender does not match the configured cable type.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Configure the robot cable extender with the correct cable type, (B) Connect the intended cable between the cable extenders (C) Update Software, (D) Contact your local Universal Robots service provider for assistance

C351A10 Detected cable type is {unsigned}

C351A11 Configured cable type is {unsigned}

C351A12 Unsupported robot for robot cable extender.

EXPLANATION

The cable extender does not support the configured robot.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Connect a supported cable type, (B) Verify correct robot configuration, (C) Contact your local Universal Robots service provider for assistance

2.67. C352 Backdrive

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C352A0 Parking timed out

EXPLANATION

The backdrive parking procedure took too long.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the robot is not vibrating due to external sources, (B) Make sure to let go of the robot when exiting back-drive, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C352A1 The joint moved more than allowed during the parking procedure

EXPLANATION

Either there is a problem with the brake system or the robot has been pushed hard enough that it moved through the friction brake.

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Make sure to let go of the robot when exiting backdrive, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C352A2 The joint moved more than allowed during the parking procedure**EXPLANATION**

Either there is a problem with the brake system or the robot has been pushed hard enough that it moved through the friction brake.

SUGGESTION

The robot brake system is in need of repair, contact your local Universal Robots service provider for assistance

C352A3 The brake solenoid was not engaged for the parking procedure. Solenoid voltage: {float}**EXPLANATION**

The brake solenoid must be engaged to perform the backdrive parking procedure.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

2.68. C353 IMU

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C353A1 IMU B selected gyro resolution not supported: {unsigned}**EXPLANATION**

Initialization of IMU B has failed during gyro configuration.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C353A2 IMU B gyro verification failed: {unsigned}**EXPLANATION**

Initialization of IMU B has failed during verification of gyro configuration.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C353A3 IMU B selected acceleration resolution not supported: {unsigned}**EXPLANATION**

Initialization of IMU B has failed during acceleration configuration.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C353A4 IMU B accelerometer verification failed: {unsigned}**EXPLANATION**

Initialization of IMU B has failed during verification of accelerometer configuration.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C353A5 Timeout occurred while configuring IMU A**EXPLANATION**

A timeout occurred while configuring inertial measurement unit A

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance, (D) Replace the BFB PCB

C353A6 Timeout occurred while configuring IMU B**EXPLANATION**

A timeout occurred while configuring inertial measurement unit B

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance, (D) Replace the BFB PCB

C353A7 Unable to detect IMU A type**EXPLANATION**

The IMU A type could not be detected

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Replace Base Filter Board (C) Contact your local Universal Robots service provider for assistance

C353A8 Unable to detect IMU B type**EXPLANATION**

The IMU B type could not be detected

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Replace Base Filter Board (C) Contact your local Universal Robots service provider for assistance

C353A9 The ISM330IS IMU data transfer failed**EXPLANATION**

The ISM330IS IMU data transfer failed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C353A10 The ICM42688 IMU data transfer failed**EXPLANATION**

The ICM42688 IMU data transfer failed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C353A11 The IMU data transfer failed for IMU A**EXPLANATION**

The IMU data transfer failed for IMU A

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C353A12 The IMU data transfer failed for IMU B

**EXPLANATION**

The IMU data transfer failed for IMU B

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

2.69. C354 Initialization Error

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C354A1 The SCB has entered the fault state

EXPLANATION

The SCB faulted during the robot initialization

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A2 The SCB has entered the violation state

EXPLANATION

The SCB entered the violation state during the robot initialization

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A3 The SCB did not reach the correct state

EXPLANATION

The SCB did not reach the correct state during the robot initialization

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A4 Failed to send the safety configuration

EXPLANATION

Unable to send the safety configuration during the robot initialization

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A5 Error during boot status validation

EXPLANATION

The device did not reach a responsive state after performing a reboot

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A6 Powering down the robot arm took too long

EXPLANATION

It took too long to power down the robot arm which is required before a reboot

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A7 Failed to close communication devices

EXPLANATION

Closing all communication devices is required before performing a reboot of the SCB FPGA

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A8 Failed to open communication devices**EXPLANATION**

Unable to re-establish connection with communication devices

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A9 Failed to hotplug FPGA devices**EXPLANATION**

The FPGA devices must be hotplug removed before loading drivers

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A10 Failed to unload FPGA drivers**EXPLANATION**

The FPGA devices must unload the drivers before new ones can be loaded

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A11 Failed to load FPGA drivers**EXPLANATION**

The FPGA devices must load the new drivers before continuing the reboot process

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A12 The Teach Pendant does not respond**EXPLANATION**

The Teach Pendant was rebooted but did not reach a responsive state

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A13 The Injection-Molding-Machine-Interface does not respond**EXPLANATION**

The Injection-Molding-Machine-Interface was rebooted but did not reach a responsive state

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A14 The Injection-Molding-Machine-Interface_FPGA does not respond**EXPLANATION**

The Injection-Molding-Machine-Interface_FPGA was rebooted but did not reach a responsive state

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance



C354A15 The SCB does not respond

EXPLANATION

The SCB was rebooted but did not reach a responsive state

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A16 Failed to start Xillybus device scanning

EXPLANATION

Unable to scan for Xillybus devices which is required before loading the drivers

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C354A17 Failed to read the safety configuration

EXPLANATION

Unable to read the safety configuration during the robot initialization

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reload the installation, (B) Make new installation, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C354A18 Communication initialization failed

EXPLANATION

Failed to initialize communication to the SCB FPGA during the reboot process

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

2.70. C355 Safety API

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C355A1 Unknown safety system message, id {unsigned}

EXPLANATION

The safety API client message is unknown to the UR safety system.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C355A2 Unknown safety system version, id {unsigned}

EXPLANATION

The safety API client message version is unknown to the UR safety system.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C355A4 Violation caused by safety API client

EXPLANATION

Safety API client violation request

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Contact the provider of the safety API client

C355A5 Safeguard reset time is out of range, received value {unsigned}ms**EXPLANATION**

The supported range is 0ms to 10000ms (10s)

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Contact the provider of the safety API client

C355A6 Retransmission of configuration is not allowed**EXPLANATION**

The safety system received more than one configuration message

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Contact the provider of the safety API client

C355A7 Provided an invalid configuration**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Contact the provider of the safety API client

C355A8 The execution time is greater than the period for supervisor with ID {unsigned}**C355A9 The supervisor max execution time is {unsigned}ms****C355A10 The max supervisor period is {unsigned}ms****C355A11 Restart period is outside allowed range for supervisor with ID {unsigned}****C355A12 The applied value is {unsigned}****C355A13 The upper limit is {unsigned}****C355A14 The lower limit is {unsigned}****C355A15 Supervisor period is outside allowed range for supervisor with ID {unsigned}****C355A16 Supervisor max execution time is outside allowed range for supervisor with ID {unsigned}****C355A17 Hashes are identical for supervisor with ID {unsigned}****C355A18 Seeds are identical for supervisor with ID {unsigned}****C355A19 Supervisor with ID {unsigned} is faster than the limitset supervisor****C355A20 The limitset supervisor period is {unsigned}ms****C355A21 The actual supervisor period is {unsigned}ms****C355A22 The limitset supervisor is not configured****C355A23 Timeout violation caused by supervisor with ID {unsigned}****C355A24 Time passed {unsigned}ms****C355A25 Validation of hash failed for supervisor with ID {unsigned}****C355A26 Actual seed {hex}****C355A27 Expected hash {hex}****C355A28 Actual hash {hex}****C355A29 Alternation of seed failed for supervisor with ID {unsigned}**



C355A30 Incompatible Safety URCap

EXPLANATION

The Safety URCap is not compatible with this version of software

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact the provider of the safety API client

C355A31 Failed UUID seed validation

EXPLANATION

The Safety URCap failed to send a valid seed for identification

SUGGESTION

Try the following actions to see which resolves the issue: (A) Uninstall software safety equipment, (B) Update software, (c) Contact the provider of the safety API client

C355A32 Invalid tool speed limit, value was {float}

EXPLANATION

The safety system only accepts rational numbers.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact the provider of the safety API client

C355A33 Invalid stopping distance limit, value was {float}

EXPLANATION

The safety system only accepts rational numbers.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact the provider of the safety API client

C355A34 The received hash does not match the hash from the installation

EXPLANATION

The hash from the installation must match the received hash

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Check that safety plugin installation node is configured correctly, (C) Contact the provider of the safety API client

C355A35 The identifier is incorrect

EXPLANATION

The identifier used by the client does not match with the expected identifier

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact the provider of the safety API client

C355A36 Validation of crc failed for message ID {unsigned}

EXPLANATION

The calculated checksum of the message does not match the received checksum

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact the provider of the safety API client

2.71. C357 Too high static load on {deviceName}

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C357A0 Payload CoG is outside reach of robot

EXPLANATION

The robot is not able to lift this much payload in its current position

SUGGESTION

The payload CoG is extended too far out horizontally from the base of the robot.

C357A1 Reorientation of tool causes load on joint to be too high

EXPLANATION

The robot is not able to lift this much payload in its current position

SUGGESTION

The payload CoG causes too high load on the joints. Please re-orient the wrist to move the payload CoG closer to the rotation axes of the joints, e.g. horizontally downwards.

2.72. C359 Report system

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C359A0 Missing violation handler

EXPLANATION

A violation was reported but no violation handler was registered

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C359A1 Missing critical fault handler

EXPLANATION

A critical fault was reported but no critical fault handler was registered

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

2.73. C361 Gear torsion

C361A0 Excessive gear torsion registered

EXPLANATION

The joint has previously measured an excessive gear torsion, thus the gear/encoders may be damaged

SUGGESTION

Contact your local Universal Robots service provider for assistance

C361A1 State transition is not permitted following an excessive gear torsion

EXPLANATION

The joint has previously measured an excessive gear torsion, thus it cannot be safely brake released

SUGGESTION

Contact your local Universal Robots service provider for assistance

2.74. C362 State Machine

C362A1 Unexpected state {unsigned} encountered

**EXPLANATION**

The FSM is trying to enter unexpected state

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C362A2 Registration of state {unsigned} have failed**EXPLANATION**

The FSM is trying to register a state, but failed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.75. C363 Tool Accelerometer

C363A0 Unexpected identifier {hex} read from the accelerometer**EXPLANATION**

The identifier value read from the accelerometer was different from the expected value

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace Tool PCBA, (D) Contact your local Universal Robots service provider for assistance

C363A1 The configuration that was read back from the accelerometer does not match the expected configuration**EXPLANATION**

The configuration that was read back from the accelerometer after initialization was not as expected

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace Tool PCBA, (D) Contact your local Universal Robots service provider for assistance

C363A2 Failed to write to a configuration register with data {hex}**EXPLANATION**

The accelerometer configuration register could not be written to

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace Tool PCBA, (D) Contact your local Universal Robots service provider for assistance

2.76. C364 Robot validation

Description

When powering on the robot arm, prior to releasing the brakes, the robot type stored in the arm is validated against the robot configuration, and the arm will be prevented from releasing the brakes if the types do not match.

If there is a mismatch between the stored and configured type, it will result in the following error when attempting to brake release:

[2.76 C364 Robot validation on the previous page](#)

C364A0: Robot validation : Robot type mismatch between configuration and arm

C364A1: Robot validation : Robot type disagreement in the joints.

The error indicates that the control box is configured for a different robot type than the attached arm. If the SD-card has been moved between different control boxes, or the control box used for the robot arm has been changed, this can be the cause for the configuration mismatch.

Correct the issue by changing back to the original SD-card or control box.

If these are not available, a clean SD-card image downloaded from the UR Support Site can be flashed to the SD card, and the correct configuration will be used after entering the robot serial number on the serial number screen.

C364A0 Robot type mismatch between configuration and arm

EXPLANATION

The robot type specified in the configuration does not match the type stored in the robot arm

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Ensure that the robot arm match the control box configuration, (C) Update software, (D) Consult the Universal Robots service manual, (E) Contact your local Universal Robots service provider for assistance

C364A1 Robot type disagreement in the joints

EXPLANATION

The robot type stored in the joints do not match. This can happen if a joint has been replaced with one from a different robot type

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Consult service manual and reupload the robot type to all joints, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C364A2 The robot type stored in the joints is {unsigned}

C364A3 The robot type in the configuration is {unsigned}

2.77. C365 Teach Pendant RS485 communication

C365A0 Critical error

EXPLANATION

A critical error occurred in the TP RS485 driver

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.78. C366 Tool panel interface



C366A0 The panel interface output voltage of {float}V is too high while the voltage was not turned on

EXPLANATION

The panel interface output voltage is too high while it is supposed to be off

C366A1 The panel interface output voltage of {float}V is lower than allowed

EXPLANATION

The panel interface output voltage is lower than allowed

C366A2 The panel interface output voltage of {float}V is higher than allowed

EXPLANATION

The panel interface output voltage is higher than allowed

C366A3 A fault event was detected on supply to the panel interface

EXPLANATION

A fault event can be caused by voltage, current or temperature issues on the supply to the panel interface

C366A4 An over current fault event was detected on panel interface digital outputs

EXPLANATION

A fault event is caused by higher current being detected on one of the digital outputs than specified for the panel interface

2.79. C367 Power Control Board

C367A0 Failed to initialize the robot power sensor

EXPLANATION

The robot power sensor initialization failed

C367A1 Failed to initialize the ERD power sensor

EXPLANATION

The ERD power sensor initialization failed

C367A2 Failed to initialize the voltage monitor sensor

EXPLANATION

The voltage monitor sensor initialization failed

C367A3 Failed to read data from the robot power sensor

EXPLANATION

The robot power sensor read failed

C367A4 Failed to read data from the ERD power sensor

EXPLANATION

The ERD power sensor read failed

C367A5 Failed to read data from the voltage monitor sensor

EXPLANATION

The voltage monitor sensor read failed

C367A6 Power Control Board revision invalid

EXPLANATION

The Power Control Board revision is not valid

C367A7 Power Control Board revision {unsigned} is not supported

EXPLANATION

The Power Control Board revision is not supported

C367A9 The voltage monitor chip is not responding

EXPLANATION

The voltage monitor chip did not respond during initialization

2.80. C368 Power Supply Unit

C368A0 Failed to communicate with power supply unit

EXPLANATION

The I2C communication with the power supply unit timed out or returned an error

2.81. C370 Ethernet MDIO

C370A0 When communicating with a an ethernet PHY, the command has timed out. command: {hex}

EXPLANATION

An issue has been detected communicating with an Ethernet PHY over the MDIO interface.

2.82. C371 End of Arm Ethernet

C371A0 Unable to read status from the End of Arm Ethernet

EXPLANATION

The End of Arm Ethernet is not responding to status requests

C371A3 Unable to establish link to the Control Box

EXPLANATION

The End of Arm Ethernet never established a successful link to the Control Box

C371A4 Unable to establish link to the Tool

EXPLANATION

The End of Arm Ethernet never established a successful link to the Tool

C371A5 Validation of PHY device with ID: {hex} failed

EXPLANATION

Failed during validation of the connected ethernet PHY device

2.83. C373 Power Interface Board

C373A0 Failed to initialize the monitor sensors

EXPLANATION

The monitor sensors initialization failed

C373A1 Failed to initialize the battery interface sensor

EXPLANATION

The battery interface sensor initialization failed

C373A2 Failed to read data from the battery interface sensor

EXPLANATION

The battery interface sensor read failed



C373A3 Failed to read data from the monitor sensors

EXPLANATION

The monitor sensors read failed

C373A4 Power Interface Board revision invalid

EXPLANATION

The Power Interface Board revision is not valid

C373A5 Power Interface Board revision {unsigned} is not supported

EXPLANATION

The Power Interface Board revision is not supported

2.84. C375 Motor encoder

C375A0 Critical error

EXPLANATION

A critical communication error occurred

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR service provider for assistance

2.85. C377 Charging system

C377A0 The motors are active while charging is enabled

EXPLANATION

The motors are active while charging is enabled, which is not allowed

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR service provider for assistance

2.86. C400 Elbow position close to safety plane limits

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

2.87. C401 Exceeding user safety settings for stopping time

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

2.88. C402 Exceeding user safety settings for stopping distance

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

2.89. C403 Danger of clamping between the lower arm and tool flange of the robot

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

2.90. C404 Unexpected behavior

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C404A0 Runtime sends data too often

C404A1 Runtime tries to receive data too often

2.91. C450 Force-Torque sensor

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C450A0 Sensor data invalid

EXPLANATION

Force-Torque sensor is defective or not mounted correctly

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for loose connections, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C450A1 Sensor can not be used, therefore it is disabled

EXPLANATION

Force-Torque sensor version is newer than the Robot software

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for loose connections, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C450A4 Force-Torque sensor is expected, but it cannot be detected

EXPLANATION

Force-Torque sensor is expected, but no signals from the sensor can be detected.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Check for loose connections, (C) Check for damages to the Tool/sensor, (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C450A5 Force-Torque sensor is detected but not calibrated

EXPLANATION

Force-Torque sensor is installed, but no calibration was found.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Contact your local Universal Robots service provider for assistance



2.92. C451 Tool Power Sensor

C451A5 Failed to initialize the M8 GND sensor chip

EXPLANATION

The tool M8 GND sensor chip is not responding correctly.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the tool PCB, (D) Contact your local Universal Robots service provider for assistance

C451A6 Failed to initialize the M8 digital output low side sensor chip

EXPLANATION

The tool M8 digital output low side sensor chip is not responding correctly.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the tool PCB, (D) Contact your local Universal Robots service provider for assistance

C451A8 Too many failing M8 GND sensor readings in a row

EXPLANATION

The tool M8 GND sensor chip is not responding correctly.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the tool PCB, (D) Contact your local Universal Robots service provider for assistance

C451A9 Too many failing M8 digital output low side sensor readings in a row

EXPLANATION

The tool M8 digital output low side sensor chip is not responding correctly.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the tool PCB, (D) Contact your local Universal Robots service provider for assistance

C451A11 Failed to initialize the tool power sensor chip

EXPLANATION

The tool power sensor chip is not responding correctly.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the tool PCB, (D) Contact your local Universal Robots service provider for assistance

C451A13 Too many failing power sensor readings in a row

EXPLANATION

The tool power sensor chip is not responding correctly.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the tool PCB, (D) Contact your local Universal Robots service provider for assistance

2.93. C453 SCB IMU

C453A2 Unable to validate the IMU type, received type {hex}

EXPLANATION

The IMU is not recognized.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the Safety Control Board, (D) Contact your local MiR service provider for assistance

C453A3 Unable to initialize the IMU**EXPLANATION**

The IMU is not responding correctly

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the Safety Control Board, (D) Contact your local MiR service provider for assistance

C453A4 Timeout while waiting for new IMU data**EXPLANATION**

The IMU has stopped providing new data

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the Safety Control Board, (D) Contact your local MiR service provider for assistance

C453A5 Failed to start SPI transfer for the IMU**EXPLANATION**

The SPI transfer for the IMU did not start correctly

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local MiR service provider for assistance

2.94. C500 Self-test step

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C500A19 Awaiting acceptance started**EXPLANATION**

The Self-test awaits acceptance using serial number.

2.95. C526 Node ID

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C526A0 Index/node id {unsigned} is not within the range for a joint.**EXPLANATION**

The index/node id provided is not in the range for a valid joint.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C526A1 Node {unsigned} is not a joint.**EXPLANATION**

The provided node is not a joint.

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.96. C710 ROM Test

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C710A0 Critical error

EXPLANATION

A critical error occurred during ROM validation

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.97. C720 LVD (low voltage detection)

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C720A6 The VCCINT voltage of {float} is too high

EXPLANATION

The internal VCCINT voltage is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the PCBA, (D) Contact your local Universal Robots service provider for assistance

C720A7 The VCCINT voltage of {float} is too low

EXPLANATION

The internal VCCINT voltage is too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the PCBA, (D) Contact your local Universal Robots service provider for assistance

C720A8 The VCCAUX voltage of {float} is too high

EXPLANATION

The internal VCCAUX voltage is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the PCBA, (D) Contact your local Universal Robots service provider for assistance

C720A9 The VCCAUX voltage of {float} is too low

EXPLANATION

The internal VCCAUX voltage is too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the PCBA, (D) Contact your local Universal Robots service provider for assistance

C720A10 The VCCBRAM voltage of {float} is too high

EXPLANATION

The internal VCCBRAM voltage is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the PCBA, (D) Contact your local Universal Robots service provider for assistance

C720A11 The VCCBRAM voltage of {float} is too low**EXPLANATION**

The internal VCCBRAM voltage is too low

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the PCBA, (D) Contact your local Universal Robots service provider for assistance

2.98. C740 Hardware monitoring

You may find the following links helpful in addressing this issue:

- [How to perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C740A0 Critical error**EXPLANATION**

A critical error occurred during hardware monitoring

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A10 24V IO voltage is outside of the allowed range: {float}V**EXPLANATION**

An error occurred in the 24V supply

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check for shorts on anything attached to the IOs, (B) Conduct a complete rebooting sequence, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C740A11 48V voltage is outside of the allowed range: {float}**EXPLANATION**

The 48V supply voltage is outside the allowed range. This may be due to the robot drawing too much current.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Adjust the robot program to reduce peak acceleration and torques, (B) Ensure that there is no electrical short in the Control Box or robot, (C) Conduct a complete rebooting sequence (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C740A15 The {float}A current draw of the robot is outside the allowed range.**EXPLANATION**

The current draw of the robot is higher than what is supported by the power supply.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Adjust the robot program to reduce peak acceleration and torques, (B) Ensure that the robot power cable is not shorted during operation, (C) Conduct a complete rebooting sequence (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C740A20 24V IO voltage is outside of the allowed range: {float}V

**EXPLANATION**

An error occurred in the 24V IO supply

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the fuse in the control box, (B) Check supply to the IOs is attached, (C) Check for shorts on anything attached to the IOs, (D) Conduct a complete rebooting sequence, (E) Update software, (F) Contact your local Universal Robots service provider for assistance

C740A21 24V IO current is outside of the allowed range: {float}A**EXPLANATION**

Too high current draw on 24V IO supply

SUGGESTION

Check 24V IO connections to make sure the IO current are within specified limits

C740A24 The left Three-Position Enabling button is inconsistent**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Do not press the button repeatedly, (B) Remove any pressure off the button, (C) Press the button with more pressure. Light pressing may not work, (D) Update software, (E) contact your local Universal Robots service provider for assistance.

C740A25 The right Three-Position Enabling button is inconsistent**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Do not press the button repeatedly, (B) Remove any pressure off the button, (C) Press the button with more pressure. Light pressing may not work, (D) Update software, (E) contact your local Universal Robots service provider for assistance.

C740A28 uA's 3V3voltage ADC ref is outside of the allowed range: {float}**EXPLANATION**

The ADC reference voltage for uA's 3V3 supply is outside of the allowed range, which may cause inaccurate voltage measurements.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the PCBA, (D) Contact your local Universal Robots service provider for assistance

C740A29 The solenoid driver curcuit encountered an error condition**EXPLANATION**

The fault signal from the solenoid driver curcuit was asserted, indicating an issue with the joint electronics.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Replace the joint PCB.

C740A30 The brake solenoid could not be detected**EXPLANATION**

The joint was unable to detect a connected brake solenoid

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Verify that the brake solenoid cable is connected to the joint PCB, (C) Verify that the brake solenoid is not damaged.

C740A31 The robot current offset limit check has failed**EXPLANATION**

The robot current offset value is outside of the allowed range

SUGGESTION

Try the following actions to resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance.

C740A32 The robot current offset integrity check has failed

EXPLANATION

The robot current offset value can't be trusted as the integrity check has failed

SUGGESTION

Try the following actions to resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance.

C740A33 48V voltage is below the minimum allowed limit: {float}**EXPLANATION**

The 48V supply voltage is below the minimum allowed limit.

SUGGESTION

Try the following actions to see which resolve the issue: (A) Make sure that the combination of robot arm and control box is supported, (B) Adjust the robot program to reduce peak acceleration and torques, (C) Ensure that there is no electrical short in the Control Box or robot, (D) Conduct a complete rebooting sequence (E) Update software, (F) Contact your local Universal Robots service provider for assistance

C740A34 48V voltage is above the maximum allowed limit: {float}**EXPLANATION**

The 48V supply voltage is above the maximum allowed limit.

SUGGESTION

Try the following actions to see which resolve the issue: (A) Make sure that the combination of robot arm and control box is supported, (B) Adjust the robot program to reduce peak decelerations and torques, (C) Ensure that there is no electrical short or loose connections in the Control Box or robot, (D) Conduct a complete rebooting sequence (E) Update software, (F) Contact your local Universal Robots service provider for assistance

C740A36 The M8 IO current offset measurement of {float} A is outside of the allowed range.**EXPLANATION**

The M8 IO current offset measurement is outside of the allowed range.

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Replace the tool PCBA, (C) Contact your local Universal Robots service provider for assistance

C740A38 48V Safety Control Board voltage is below the minimum allowed limit: {float}**EXPLANATION**

The 48V Safety Control Board supply voltage is below the minimum allowed limit.

SUGGESTION

Try the following actions to see which resolve the issue: (A) Adjust the robot program to reduce peak acceleration and torques, (B) Ensure that there is no electrical short or loose connections in the Control Box or robot, (C) Conduct a complete rebooting sequence (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C740A39 48V Safety Control Board voltage is above the maximum allowed limit: {float}**EXPLANATION**

The Safety Control Board SCB supply voltage is above the maximum allowed limit.

SUGGESTION

Try the following actions to see which resolve the issue: (A) Adjust the robot program to reduce peak decelerations and torques, (B) Ensure that there is no electrical short or loose connections in the Control Box or robot, (C) Conduct a complete rebooting sequence (D) Update software, (E) Contact your local Universal Robots service provider for assistance

C740A40 5V1 voltage is outside of the allowed range: {float}**C740A41 12VA voltage on the Power Control Board is outside of the allowed range: {float}****C740A42 5VA voltage on the Power Control Board is outside of the allowed range: {float}**



C740A43 3V3A voltage on the Power Control Board is outside of the allowed range: {float}

C740A44 2V5 reference voltage on the Power Control Board is outside of the allowed range: {float}

C740A45 PSU 48V voltage is outside of the allowed range: {float} V

EXPLANATION

The 48V voltage reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A46 PSU 48V current is outside of the allowed range: {float} A

EXPLANATION

The 48V current reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A47 PSU 24V voltage is outside of the allowed range: {float} V

EXPLANATION

The 24V voltage reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A48 The 24V current reported by the PSU is outside of the allowed range: {float} A

EXPLANATION

The 24V current reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A49 PSU 12V voltage is outside of the allowed range: {float} V

EXPLANATION

The 12V voltage reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A50 PSU AC voltage is outside of the allowed range: {float} V

EXPLANATION

The mains AC voltage reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A51 PSU AC current is outside of the allowed range: {float} A

EXPLANATION

The mains AC current reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A52 PSU AC power is outside of the allowed range: {float} W

EXPLANATION

The mains AC power reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A53 PSU temperature is outside of the allowed range: {float} degrees C

EXPLANATION

The hot-spot temperature reported by the PSU is outside of the allowed range

SUGGESTION

Try the following actions to see which resolve the issue: (A) Ensure that the fan on the control box is running and air flow is not obstructed, (B) Conduct a complete rebooting sequence (C) Update software, (D) Contact your local Universal Robots service provider for assistance

C740A54 The PCB type: {unsigned} is unsupported by the SW Version.

EXPLANATION

The hardware is not supported by this software version

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

C740A55 Vin on the Power Interface Board is outside of the allowed range: {float} V

EXPLANATION

The input voltage on the Power Interface Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the battery, (B) Ensure proper connections, (C) Replace the Power Interface Board, (D) Contact your local MiR service provider for assistance

C740A56 5V on the Power Interface Board is outside of the allowed range: {float} V

EXPLANATION

The 5V voltage on the Power Interface Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Replace the Power Interface Board, (B) Contact your local MiR service provider for assistance

C740A57 5VA on the Power Interface Board is outside of the allowed range: {float} V

EXPLANATION

The 5VA voltage on the Power Interface Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Replace the Power Interface Board, (B) Contact your local MiR service provider for assistance

C740A58 3V3A on the Power Interface Board is outside of the allowed range: {float} V

EXPLANATION

The 3V3A voltage on the Power Interface Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Replace the Power Interface Board, (B) Contact your local MiR service provider for assistance

C740A59 2V5 reference on the Power Interface Board is outside of the allowed range: {float} V

EXPLANATION

The 2V5 reference voltage on the Power Interface Board is outside of the allowed range

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Replace the Power Interface Board, (B) Contact your local MiR service provider for assistance

C740A60 2V5 ISO reference on the Power Interface Board is outside of the allowed range: {float} V**EXPLANATION**

The 2V5 ISO reference voltage on the Power Interface Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Replace the Power Interface Board, (B) Contact your local MiR service provider for assistance

C740A61 24V ISO on the Power Interface Board is outside of the allowed range: {float} V**EXPLANATION**

The 24V ISO voltage on the Power Interface Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Replace the Power Interface Board, (B) Contact your local MiR service provider for assistance

C740A62 Battery interface voltage on the Power Interface Board is outside of the allowed range: {float} V**EXPLANATION**

The battery interface voltage on the Power Interface Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Check the battery, (B) Ensure proper connections, (C) Replace the Power Interface Board, (D) Contact your local MiR service provider for assistance

C740A63 12VB voltage on the Power Control Board is outside of the allowed range: {float}**EXPLANATION**

The 12V supply voltage for uB on the Power Control Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A64 5VB voltage on the Power Control Board is outside of the allowed range: {float}**EXPLANATION**

The 5V supply voltage for uB on the Power Control Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A65 3V3B voltage on the Power Control Board is outside of the allowed range: {float}**EXPLANATION**

The 3.3V supply voltage for uB on the Power Control Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C740A66 48V voltage on the Power Control Board is outside of the allowed range: {float}

EXPLANATION

The 48V supply voltage on the Power Control Board is outside of the allowed range

SUGGESTION

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

2.99. C741 Hardware general information

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C741A2 The hardware is not supported by this software version

EXPLANATION

The hardware is not supported by this software version

SUGGESTION

Try the following actions to see which resolves the issue: (A) Update software, (B) Contact your local Universal Robots service provider for assistance

2.100. C742 Control Box temperature

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C742A1 The temperature of {signed} °C is close to the limit

EXPLANATION

The temperature in the Control Box is close to the limit

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce the ambient temperature, (B) Improve the ventilation around the Control Box, (C) Contact your local Universal Robots service provider for assistance

C742A2 The temperature of {signed} °C is above the limit

EXPLANATION

The temperature in the Control Box is too high

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce the ambient temperature, (B) Improve the ventilation around the Control Box, (C) Contact your local Universal Robots service provider for assistance

2.101. C743 External Axes

If you unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C743A0 Velocity limit for the external axis at index {unsigned} was exceeded

EXPLANATION

The velocity that the axis needs to complete its trajectory at the same time as the robot exceeds its maximum limit.

SUGGESTION

Lower the required axis velocity by reducing the Polyscope speed scale slider or adjust the program to reduce the robot velocity.

C743A1 The servo drive for the external axis at index {unsigned} has entered a fault state

EXPLANATION

An external axis has had a problem where the servo drive has entered a fault state

**SUGGESTION**

Check the external axis's servo drive fault indicators for vendor-specific diagnostics

C743A2 Freedrive is incompatible with external axis motion**EXPLANATION**

External axes cannot be moving while the controller is in Freedrive.

SUGGESTION

Verify that external axes are completely stopped before entering Freedrive and ensure no axis motion is commanded when already in Freedrive mode.

C743A3 The servo drive for the external axis at index {unsigned} has become disabled**EXPLANATION**

The servo drive of an enabled external axis is reporting itself as disabled

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reset drive emergency stop, (B) Check whether drive has been configured properly

C743A4 External axis homing is incompatible with other motion**EXPLANATION**

Other external axis or Robot motion was detected during an external axis servo drive homing operation

SUGGESTION

Adjust the robot program to not perform other external axis or robot motions until the external axis servo drive homing operation has finished

C743A5 Only one external axis servo drive homing operation at a time**EXPLANATION**

Two or more external axis servo drive homing operations were detected at the same time

SUGGESTION

Adjust the robot program to only perform one external axis servo drive homing operation at a time

C743A6 Motion detected within multiple external axis groups**EXPLANATION**

Only one external axis group may be moving at any given time.

SUGGESTION

Ensure all axes within an axis group are stopped before attempting to move axes within another group.

C743A7 The position of external axis at index {unsigned} is close to joint limit**EXPLANATION**

The position that the axis needs to complete its trajectory at the same time is close to its joint limit.

C743A8 The position of external axis at index {unsigned} is close to upper joint limit**EXPLANATION**

The position that the axis needs to complete its trajectory at the same time is close to its upper joint limit.

SUGGESTION

Change the direction of target velocity of the axis so that it is going to move back into the safe zone in next step.

C743A9 The position of external axis at index {unsigned} is close to lower joint limit**EXPLANATION**

The position that the axis needs to complete its trajectory at the same time is close to its lower joint limit.

SUGGESTION

Change the direction of target velocity of the axis so that it is going to move back into the safe zone in next step.

C743A10 The position of external axis at index {unsigned} was commanded to move without the axis enabled.

EXPLANATION

An axis must be enabled before it can be moved.

SUGGESTION

Enable the axis and try again.

2.102. C746 Cable Extender

You may find the following links helpful in addressing this issue:

- How to [perform a complete reboot](#)
- [Find a distributor](#) page

If you are unable to resolve the issue, log in to <http://myUR.universal-robots.com> and create a new case.

C746A0 The house temperature is too low: {float} Celsius

EXPLANATION

The house temperature is below the allowed limit

SUGGESTION

Try the following actions to see which resolves the issue: (A) Increase the ambient temperature, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C746A1 The house temperature is too high: {float} Celsius

EXPLANATION

The house temperature is above the allowed limit

SUGGESTION

Try the following actions to see which resolves the issue: (A) Reduce the ambient temperature, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C746A2 The energy eater temperature is too low: {float} Celsius

EXPLANATION

The energy eater temperature is below the allowed limit

SUGGESTION

Try the following actions to see which resolves the issue: (A) Increase the ambient temperature, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C746A3 The energy eater temperature is too high: {float} Celsius

EXPLANATION

The energy eater temperature is above the allowed limit

SUGGESTION

Try the following actions to see which resolves the issue: (A) Decrease the robot accelerations and speeds, (B) Update software, (C) Reduce the ambient temperature, (D) Contact your local Universal Robots service provider for assistance

C746A4 Cable type mismatch between near and far cable extenders

EXPLANATION

The cable extenders does not have the same cable type

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure that the robot cable extender is configured properly, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C746A5 The energy eater regulation is inactive

EXPLANATION

The energy eater regulation is not running at the expected frequency

**SUGGESTION**

Try the following actions to see which resolves the issue: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C746A7 The connected cable was not recognized**EXPLANATION**

The connected cable did not have a recognized ID resistor

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure the cable is connected properly, (B) Make sure that the cable ID resistor is connected properly to the extender board, (C) Update software, (D) Contact your local Universal Robots service provider for assistance

2.103. C748 Security

C748A0 Library error: {unsigned}**EXPLANATION**

The device has failed security checks

SUGGESTION

Try the following actions to see which resolves the issue: (A) Make sure to use official software provided by UR, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C748A3 Elevation aborted (bootloader is not capable)**EXPLANATION**

The device cannot be provisioned, because the bootloader lacks the required features

SUGGESTION

Make sure to program a compatible bootloader

C748A4 Elevation aborted (key revision {unsigned} is not valid)**EXPLANATION**

The device cannot be provisioned with that key revision

SUGGESTION

Make sure to send only valid key revisions in the provisioning command

C748A5 Elevation aborted (key revision {unsigned} is not supported with a non-red bootloader)**EXPLANATION**

The device cannot be provisioned with development bootloaders

SUGGESTION

The key revision used is not allowed with the existing color of the bootloader

2.104. C749 Digital output

C749A1 Excessive load**EXPLANATION**

The load current exceeded safe limits, or an inductive load issue was detected

SUGGESTION

Try the following actions: (A) Reduce the load current and let the system cool, (B) For load currents above 600mA, the inductive clamping energy must be limited to 0.3 Joule, (C) Contact your local Universal Robots service provider for assistance

C749A8 Internal hardware fault

EXPLANATION

A critical internal fault was detected on the digital output device

SUGGESTION

Try the following actions: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C749A27 Failed to start communication with digital output device {unsigned}**EXPLANATION**

An internal communication failure occurred while initializing the device

SUGGESTION

Try the following actions: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C749A28 Unexpected response size from digital output device {unsigned}**EXPLANATION**

An unexpected response was received while initializing the device

SUGGESTION

Try the following actions: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C749A29 Invalid CRC from digital output device {unsigned}**EXPLANATION**

A communication integrity check failed while initializing the digital output device

SUGGESTION

Try the following actions: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Replace the safety control board, (D) Contact your local Universal Robots service provider for assistance

C749A30 Could not update digital output state**EXPLANATION**

An internal communication conflict prevented the digital output state from being updated

SUGGESTION

Try the following actions: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C749A31 Failed to update digital output state**EXPLANATION**

An internal communication failure prevented the digital output state from being updated

SUGGESTION

Try the following actions: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

C749A32 Unexpected transfer size {unsigned} when updating digital output**EXPLANATION**

An internal communication issue occurred while updating the output state

SUGGESTION

Try the following actions: (A) Conduct a complete rebooting sequence, (B) Update software, (C) Contact your local Universal Robots service provider for assistance

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